Social Security Design and Its Political Support*

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Abstract

The design of social security systems has implications for the political support of intergenerational transfers. We examine the effects of contributive vs. universal pensions, as well as the link between benefits and earnings, on the social security tax rate, labor supply, and capital accumulation. In an overlapping generations model featuring heterogeneous households, our results align with existing ones on the effects of retirees' relative political power, ageing, and income inequality. Additionally, we find taxes to be lower when benefits are earnings-related and, if informal retirees are poorer than formal ones and the pension a comparable share of their consumption, higher when pensions are universal. Calibrated to Argentina, the model accounts for about three quarters of the rise in pension spending after the 2005–2010 expansion of coverage. In the model calibrated to the U.S., France and the UK, ageing is the main determinant of pension spending, the earnings link the second, and income inequality and voting patterns minor. When the electorate also chooses the design of the system, an interior choice requires redistribution through pensions to be costly. With a deadweight cost of flat benefits that rises with the size of the system, calibrated to the U.S., ageing makes benefits more earnings-related, while income inequality moves the design in a direction that depends on the intertemporal elasticity of substitution, in line with the OECD cross-section, where the earnings link is stronger in older populations and unrelated to inequality.

JEL Classification: D72, H55, J22, O17

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1 Introduction

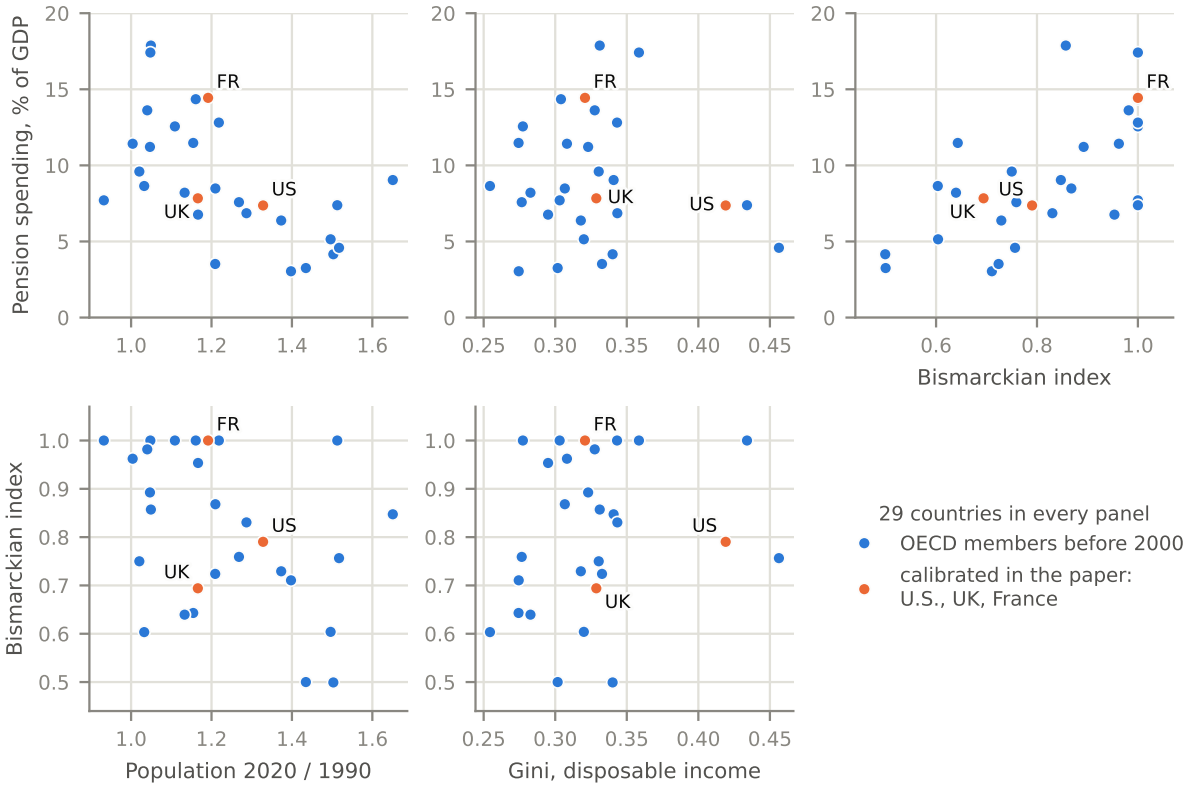
Pensions are the largest transfer programme of most governments. In 2020 public pension spending averaged about 8% of GDP across the OECD and close to 4% in emerging and middle-income economies.[1] Systems differ along two key dimensions. The first concerns eligibility and coverage: some condition benefits on a contribution record, others offer universal entitlements. The second is the link between benefits and past earnings: Bismarckian systems tie the two closely, whereas Beveridgean systems pay flat benefits. Both dimensions shape the labor supply of workers, the redistribution embedded in the system, and the coalition of voters that supports it.

Given their size, it is no surprise that the design of social security systems remains a central concern in policy debates. As populations age, policymakers must navigate difficult trade-offs among intergenerational equity, fiscal sustainability, and labor market efficiency. Moreover, pension arrangements are not determined by actuarial considerations alone: they are the outcome of political processes in which heterogeneous voters and organized interests shape the level and composition of benefits and contributions. A political-economy perspective is therefore critical for understanding why countries differ in the size of their systems and in their design. Figure 1.1 shows how they differ across the 29 countries that were in the OECD before 2000. Pension spending and the Bismarckian–Beveridgean index, the ratio of replacement rates at mean and at half mean earnings, both rise as population growth falls, and they rise together. Neither is related to income inequality. The last fact runs against the leading theory of pension design, that of Conde-Ruiz and Profeta (2007), in which high inequality leads a coalition of the rich and the poor to choose a small Beveridgean system and low inequality leads the middle class to choose a larger Bismarckian one, and few other politico-economic models have integrated the design. We address this gap with a dynamic politico-economic model and ask whether one political process can deliver both the size of the system and its design, and their relation to ageing but not to inequality.

We study the question in an overlapping generations model with capital accumulation and heterogeneous households. Households consume, save, and supply labor, and as voters, they choose among policies on labor income taxes and retirement benefits. Policy preferences vary by income and age; we model the resolution of these conflicts through probabilistic voting with elections taking place every period. Policy decisions not only impact economic outcomes, but also affect future policy decisions. Voters take these

[1] The dispersion is wide even among rich countries: a bit more than 3% of GDP in Ireland and Chile and more than 17% in Italy and Greece (OECD 2021).

Figure 1.1: Population growth, income inequality, pension spending and design



Sources: OECD Social Expenditure Database (pension spending), OECD historical population data (population growth), OECD, *Pensions at a Glance 2021*, table 4.1 (Bismarckian–Beveridgean index) and the UNU-WIDER World Income Inequality Database (Gini coefficient). Details are in the technical documentation (Berg and Gonzalez-Eiras 2026).

effects into account. We study Markov equilibria in which only fundamental state variables enter. Households differ in their productivity and propensity to vote. In addition, a share of households work in an informal sector or get income from home production, where labor is not taxed and savings do not enter the capital stock. Preferences are those of Greenwood et al. (1988), so that labor supply depends on the after-tax wage and how current effort raises future benefits. Benefits combine a flat component with an earnings-related one, the weight of the second being the Bismarckian index, and informal households receive a universal pension whose level is the second design parameter.

We show that when utility is logarithmic, politico-economic equilibria retain analytical tractability despite the presence of heterogeneity and institutional richness. With hand-to-mouth informal households, the equilibrium tax is a function of parameters and demographics alone. The analytical frame clarifies how population ageing, retirees' political influence, pension design, and income inequality shape the equilibrium tax rate. A more Bismarckian benefit formula lowers the equilibrium tax, since it shifts benefits from poor retirees, who value them more at the margin, to rich ones, so that retirees as a group support a smaller system. The effect of a more universal pension is ambiguous: it raises the tax when informal retirees are poorer than formal ones and the pension is

a comparable share of their consumption. When utility is constant relative risk aversion (CRRA), we show that the state space for equilibrium policies can be reduced to a single aggregate state and provide numerical solutions.[2]

We present two empirical applications of the model. First, we study Argentina as the case in which the design changed: from 2005 a series of tax amnesties raised the coverage of the elderly from 68% to over 90%, and pension spending by 1.8% of GDP. Implementing this reform in our calibrated model accounts for about three quarters of the observed rise in pension spending. The model also predicts modest declines in labor supply and savings, consistent with available evidence (Gonzalez-Rozada and Ruffo 2016). Second, initially taking the design as given, we calibrate the model to the U.S., France, and the UK, three advanced economies with distinct pension system characteristics, demographics, and inequality profiles. We find that the size of the pension system increases strongly with ageing and flat benefits and responds only little to income inequality, leisure preferences, and voting profiles.

We then let the electorate choose the earnings link as well as the tax. Without a cost of redistribution the choice is a corner. The flat component of benefits is a pure transfer within the retired cohort, from retirees above mean income to those below, and an electorate that weighs retirees by their marginal utility wants all of it: we prove that unless propensities to vote rise sharply with income, the design chosen with the tax is fully Beveridgean, whatever the tax rate, the demography and the income distribution.[3]

An interior design requires flat benefits to be more expensive than earnings-related ones. We introduce a deadweight cost of the redistribution the flat component performs, read as the loss from the implicit tax it levies on earnings (Summers 1989) and rising with the square of that tax as in Harberger (1964), so that the share of revenue lost is quadratic in the flat share of benefits, proportional to the dispersion of incomes and rising with the size of the system. A single cost parameter is calibrated so that the U.S. electorate re-elects its observed design, and it is tested on the UK and France: it accounts for much of the UK's distance from the U.S. design, though no value of it reproduces France's fully Bismarckian one. With this formulation, the model predicts that an ageing population chooses both a more Bismarckian design and higher pension spending, whereas the income distribution moves spending little and moves the design in a direction that depends on the intertemporal elasticity of substitution; this is in line with the evidence in figure 1.1.

The paper belongs to the literature on politico-economic equilibria of social security (e.g. Cooley and Soares 1999; Galasso 1999; Forni 2005; Gonzalez-Eiras and Niepelt 2008;

[2]When informal households can save, the state space includes two states.

[3]We also explore different ways of choosing the pension design, i.e. choosing the design one period in advance or choosing a permanent structure. We test the variants numerically varying parameters like the labor supply elasticity and the intertemporal elasticity of substitution, but no plausible combination avoids a corner choice for the observed U.S. level of inequality.

Song 2011), surveyed by Galasso and Profeta (2002). As in Gonzalez-Eiras and Niepelt (2008) and Song (2011), under log-GHH preferences the equilibrium has no strategic effects and taxes increase with the political power of retirees, income inequality and ageing; as in Uchida and Ono (2024), the equilibrium tax is a function of parameters and demographics alone; and as in Gonzalez-Eiras and Niepelt (2005), the effect of ageing under CRRA preferences is close to the logarithmic one after recalibration. What we add is a pension system that differs in two dimensions, in coverage and in the earnings link, and workers who differ in productivity and in their propensity to vote. Neither adds a state when informal households are hand-to-mouth, and the calibrated model quantifies what each feature of the design does to the tax rate, labor supply and savings.

On the design of the system, Galasso and Profeta (2002) leave open why some systems are Beveridgean and others Bismarckian, and why the Bismarckian ones are larger, two questions that a vote on the size of the system alone cannot answer. Casamatta et al. (2000) study the political support of Bismarckian against Beveridgean benefits, Conde-Ruiz and Profeta (2007) the joint choice of size and design, Koethenbuerger et al. (2008) the trade-off by which more redistributive systems are smaller, and Cremer et al. (2007) the choice of type and generosity. We embed the choice in a dynamic equilibrium with capital accumulation and labor supply, in which the design is chosen every period, one period ahead of the benefits it governs, and take it to the data.

Section 2 presents the model and the equilibrium concepts. Section 3 derives the equilibrium under log-GHH preferences, in which the tax is a function of parameters and demographics alone, and its comparative statics. Section 4 describes how the equilibria are computed outside that case. Section 5 calibrates the model to Argentina and compares its prediction with the observed reform, section 6 calibrates it to the U.S., France and the UK and ranks the determinants of pension spending, and section 7 lets the electorate choose the design. Section 8 concludes. The appendix collects proofs, derivations, and calibration details. An online appendix[4] collects auxiliary tables and figures and presents various robustness checks, and technical documentation accompanies the paper (Berg and Gonzalez-Eiras 2026).

2 The Model

2.1 Demographics, Inequality, and Institutions

We consider an economy populated by overlapping generations of workers and retirees. Within each generational cohort, we consider $J+1$ household types indexed $j = 0, \ldots, J$. The $j > 0$ household types allocate time between labor and leisure, pay taxes on labor income, consume, and save. When they reach retirement, they spend the return on

[4] championape.github.io/SSDclaude

their savings and any social security benefits they may receive, and die. Type $j = 0$ households similarly allocate time between labor and leisure, consume, and save for retirement. However, they supply labor to an informal labor market, thus not subject to labor income taxes, and they save for retirement using an informal savings technology. The distribution of type $j > 0$ households is given by γ_j with $\sum_{j>0} \gamma_j = 1$, and the share of type $j = 0$ households is γ_0. The ratio of workers to retirees in period t follows a deterministic process, and is given by ν_t.

The government runs a pay-as-you-go (PAYG) pension system with a balanced budget. Each period, the government imposes a tax on labor supplied in the formal sector and transfers the collected amount directly to retirees. We consider social security systems that differ along two dimensions: whether workers need to qualify to receive full benefits or not and the type of benefits provided, either tied to earnings or flat-rate. We describe the different systems in section 2.4.

Policy decisions are taken by a government that acts in the interest of voters, but lacks commitment.

2.2 Technology

In the formal sector, a continuum of competitive firms transform aggregate capital and labor into output. We define the aggregate formal labor supply as

$$H_t = N_t h_t, \qquad h_t \equiv \sum_{i>0} \gamma_i \eta_i h_{t,i}, \tag{1a}$$

where N_t is the number of formal workers at time t, $h_{t,i}$ is time allocated to labor by household type i, and η_i is their labor productivity, normalized such that $\sum_{i>0} \gamma_i \eta_i = 1$. Capital is owned by formal retirees and fully depreciates after a period. The economy-wide capital stock per formal worker, k_t, therefore corresponds to the economy-wide per capita savings of formal workers in the previous period, s_{t-1}, normalized by ν_t. We define the aggregate capital stock from

$$K_t = N_{t-1} s_{t-1}, \qquad s_{t-1} \equiv \sum_{i>0} \gamma_i s_{t-1,i}. \tag{1b}$$

The representative firm produces output using labor and capital and a conventional Cobb-Douglas technology

$$Y_t = K_t^{\alpha} H_t^{1-\alpha} \tag{2}$$

In equilibrium this implies the following factor prices:

$$R_t = \alpha \left(\frac{s_{t-1}}{\nu_t h_t} \right)^{\alpha - 1}, \qquad w_t = (1 - \alpha) \left(\frac{s_{t-1}}{\nu_t h_t} \right)^{\alpha}. \tag{3}$$

Informal households produce with a linear technology, so the informal wage w_t^0 and the return on informal savings R_t^0 are proportional to the formal sector's factor prices.

2.3 Preferences and Household Choices

Households value consumption and leisure, and discount the future at factor $\beta \in (0,1)$. Representing utility by function $u(c,h)$ with $u'_c \geq 0, u'_h \leq 0, u''_{cc} \leq 0, u''_{hh} \leq 0$, the welfare of a young household of type i born in period t is defined by

$$u\left(c_{1,t}^i, h_{t,i}\right) + \beta u\left(c_{2,t+1}^i, 0\right). \tag{4}$$

The budget constraints for formal and informal workers are given by:

$$c_{1,t}^i = w_t \eta_i h_{t,i}(1-\tau_t) - s_{t,i}, \qquad i > 0, \tag{5a}$$
$$c_{1,t}^0 = w_t^0 \eta_0 h_{t,0} - s_{t,0}, \tag{5b}$$

where τ_t is the social security tax rate. As retirees, households receive a pension b_{t+1}^i that may depend on the individual household's contribution, and thus on their labor supply, $h_{t,i}$:

$$c_{2,t+1}^i = s_{t,i} R_{t+1} + b_{t+1}^i(h_{t,i}), \qquad i > 0, \tag{5c}$$
$$c_{2,t+1}^0 = s_{t,0} R_{t+1}^0 + b_{t+1}^0(h_{t,0}). \tag{5d}$$

Optimization yields the first order conditions that represent a leisure/consumption tradeoff and a conventional Euler condition for intertemporal consumption smoothing:[5]

$$-\frac{\partial u_i}{\partial h_{t,i}} = \frac{\partial u_i}{\partial c_{1,t}^i}\left(w_t(1-\tau_t)\eta_i + \frac{\partial b_{t+1}^i/\partial h_{t,i}}{R_{t+1}}\right), \qquad i > 0 \tag{6a}$$
$$-\frac{\partial u_0}{\partial h_{t,0}} = \frac{\partial u_0}{\partial c_{1,t}^0}\left(w_t^0 \eta_0 + \frac{\partial b_{t+1}^0/\partial h_{t,0}}{R_{t+1}^0}\right), \tag{6b}$$
$$\frac{\partial u_i}{\partial c_{1,t}^i} = \beta R_{t+1} \frac{\partial u_i}{\partial c_{2,t+1}^i}, \qquad i > 0 \tag{6c}$$
$$\frac{\partial u_0}{\partial c_{1,t}^0} = \beta R_{t+1}^0 \frac{\partial u_0}{\partial c_{2,t+1}^0}. \tag{6d}$$

Optimal savings and labor supply decisions can, in general, be written as

$$h_{t,i} = h_{t,i}\left(s_{t-1}, s_t, h_t, h_{t+1}, \tau_t, \tau_{t+1}\right) \tag{7a}$$
$$s_{t,i} = s_{t,i}\left(s_{t-1}, s_t, h_t, h_{t+1}, \tau_t, \tau_{t+1}\right), \tag{7b}$$

[5] We are using shorthand notation $u_i \equiv u(c_i, h_i)$ with timing implicit by the variable we take derivatives with respect to.

where the dependence on h_{t+1}, τ_{t+1} comes from the pension system.

2.4 Social Security System

The government runs a pay-as-you-go pension system with a balanced budget. Every period, the government taxes labor income in the formal sector and pays pensions to current retirees.

We consider systems that differ along two dimensions. First, in *contributive* systems workers have to meet a contribution requirement to receive full pension benefits. In contrast, in a *universal* system all workers are eligible to receive pensions upon retirement, regardless of whether they contributed to it or not.

Second, pension schemes also differ according to the relationship between the level of contributions and benefits. *Bismarckian* systems provide earnings-related benefits; this gives workers an additional incentive to work as they perceive this increases their future pensions. In contrast, *Beveridgean* systems offer flat payments and lack this incentive. We model this by making benefits to consist of a common amount for all formal workers and a second component that is proportional to past labor income.[6]

Benefits are thus given by:

$$b_t^i = \left[\theta h_{t-1,i}\eta_i + (1-\theta)h_{t-1}\right]\bar{b}_t \tag{8a}$$

$$b_t^0 = \epsilon h_{t-1}\bar{b}_t \tag{8b}$$

where ϵ determines the level of universal pensions received by informal workers, θ measures how Bismarckian are the benefits of formal workers, and $\bar{b}_t$ is the average formal worker's pension benefits. With $\epsilon = 0$ the system is purely contributive and there are no pensions for informal workers. With $\epsilon = 1 - \theta$ they receive the basic common pension of formal workers, and with $\epsilon = 1 - \theta + \theta\eta_1 h_{t-1,1}/h_{t-1}$ the system is purely universal in the sense that informal workers receive the same pensions as the type of formal working household with lowest productivity.

The balanced budget requirement implies that

$$\nu_t w_t \tau_t h_t = \sum_{j \geq 0} \gamma_j b_t^j$$

This implies that the average pension rate $\bar{b}_t$ is:

$$\bar{b}_t = \frac{\nu_t w_t h_t \tau_t}{h_{t-1}(1 + \gamma_0 \epsilon)}. \tag{8c}$$

[6]This formulation has been frequently used in the study of redistributive social security, see e.g. Casamatta et al. (2000).

2.5 Elections

Elections take place at the beginning of each period. We assume that preferences are aggregated through probabilistic voting.[7] Thus, policy maximizes a convex combination of the objective functions of all groups of voters, where the weights reflect the groups' sizes and their responsiveness to policy changes. We allow for age related variation in responsiveness, reflected in a per capita political influence weight of unity for young voters and a per capita weight of $\omega \geq 0$ for retired voters.

2.6 Equilibrium

Competitive Equilibrium. In economic equilibrium, firms and households optimize given factor prices, policies, and a vector of aggregate states z_t, which includes exogenous demographics as well as the distribution of states $\{s_{t-1,j}, h_{t-1,j}\}_j$. In a rational expectations equilibrium, the path of policies $\{\tau_s\}_{t \leq s \leq T}$ fully determines an allocation and price system.

Conditional on some initial states z_0 and a policy sequence $\{\tau_t\}_{0 \leq t \leq T}$, a competitive equilibrium is given by an allocation and price system such that

i. households optimize;

ii. aggregate states are defined according to (1), markets clear, and factor prices are determined according to (3) for all t; and

iii. the government budget constraints in (8) are satisfied for all t.

Politico-economic equilibrium. In politico-economic equilibrium political decision makers optimally choose tax rates, taking all implications of their actions into account and forming rational expectations about future policy choices. We assume that these choices are Markov, i.e. they are functions of the fundamental state variables. The decision maker at date t takes the distribution of states $\{s_{t-1,j}, h_{t-1,j}\}$ as well as the continuation policy $\tau_t^{+1}(z_t)$ as given. Furthermore, given the continuation tax function, the policymaker takes as given the following law of motion for the state variables as a function of current policy choices:

$$z_{t+1} = \zeta_t \left(z_t, \tau_t, \tau_t^{+1} \right). \tag{9}$$

Subject to (3), (8), (9), the policymaker chooses τ_t to maximize

$$\mathcal{W}_t \left(z_t, \tau_t; \tau_t^{+1}(z_{t+1}) \right) \equiv \sum_j \gamma_j \mu_j \left[\omega \mathcal{O}_j(z_t, \tau_t) + \nu_t \mathcal{Y}_j \left(z_t, \tau_t; \tau_t^{+1}(z_{t+1}) \right) \right],$$

[7] See Lindbeck and Weibull (1987).

where the objective function is the weighted sum of indirect utilities from retirees $\mathcal{O}_j$ and workers $\mathcal{Y}_j$, with weights reflecting relative population sizes through ν_t and γ_j, relative political weight of retirees ω, and propensity to vote μ_j.

Definition 1. *A* politico-economic equilibrium *as of period t conditional on z_t consists of a sequence of tax functions, $\{\tau_\iota(\cdot)\}_{\iota\geq t}$; a sequence of continuation tax functions, $\{\tau_\iota^{+1}(\cdot)\}_{\iota\geq t}$; a sequence of laws of motion for the state variables, $\{\zeta_\iota(\cdot)\}_{\iota\geq t}$; policy choices, $\{\tau_\iota^\star\}_{\iota\geq t}$; and a competitive equilibrium allocation such that*

i. tax functions are optimal subject to continuation tax functions:

$$\tau_\iota(z_\iota) \in \arg\max_{\tau_\iota} \mathcal{W}_\iota(z_\iota, \tau_\iota; \tau_\iota^{+1}(\cdot)) \text{ for all } z_\iota, \text{ s.t. } (3), (8), (9), \ \iota \geq t;$$

ii. continuation tax functions are consistent with tax functions:[8]

$$\tau_{\iota-1}^{+1}(z_\iota) = \tau_\iota\left(z_\iota, \tau_\iota^{+1}(z_{\iota+1}(\cdot))\right) \text{ for all } z_\iota, \iota \geq t;$$

iii. laws of motion are consistent with the policy and continuation policy functions according to (9);

iv. equilibrium tax choices are generated by the continuation tax function,

$$\{\tau_\iota^\star\}_{\iota\geq t} = \tau_{\iota-1}^{+1}(z_\iota),$$

and $\{\tau_\iota^\star\}_{\iota\geq t}$ implements a competitive equilibrium allocation.

For infinite horizon and a recursive time-autonomous structure, the policy and continuation policy functions are time-autonomous functions of the state as well, and conditions i. and ii. above are combined in a fixed point requirement.

3 Logarithmic-GHH Preferences

This section derives the model's analytical results. We follow Greenwood et al. (1988) and assume

$$u(c_{1,t}^i, h_{t,i}) = \ln\left(\tilde{c}_{1,t}^i\right), \qquad \tilde{c}_{1,t}^i \equiv c_{1,t}^i - \frac{X_i \xi}{1+\xi}\left(h_{t,i}\right)^{\frac{1+\xi}{\xi}} \tag{10}$$

where ξ is the Frisch elasticity of labor supply and X_i is a type-specific taste for leisure. Proposition 1 characterizes the economic equilibrium under these log-GHH preferences. If informal households are in addition hand-to-mouth, the politico-economic equilibrium

[8] With a finite horizon, $\tau^{T+1}(z_T) = \emptyset$.

carries no state (proposition 2) and the tax rate is a function of parameters and demographics alone. The quantitative sections keep log-GHH preferences as the benchmark and let informal households save for retirement. They also report results under CRRA-GHH preferences, $u(c^i_{1,t}, h_{t,i}) = \left((\tilde{c}^i_{1,t})^{1-1/\rho} - 1\right)/(1-1/\rho)$, in which ρ is the intertemporal elasticity of substitution and $\rho = 1$ is the logarithmic case.

3.1 Economic Equilibrium

With log-GHH preferences, formal households i supply labor and save according to

$$h_{t,i} = \left(\frac{\eta_i w_t (1-\tau_t) + \frac{\theta \bar{b}_{t+1} \eta_i}{R_{t+1}}}{X_i} \right)^{\xi} \tag{11a}$$

$$s_{t,i} = \frac{\beta}{1+\beta} \left[w_t \left(1-\tau_t\right) h_{t,i} \eta_i - \frac{X_i \xi}{1+\xi} \left(h_{t,i}\right)^{\frac{1+\xi}{\xi}} \right] - \frac{b^i_{t+1}/R_{t+1}}{1+\beta}. \tag{11b}$$

Appendix A shows that in economic equilibrium every relevant variable x (savings, labor supply and consumption) is a power of capital per worker times a coefficient,

$$x_t = \Theta_{x,t} \left(\frac{s_{t-1}}{\nu_t} \right)^{\sigma_x},$$

where the coefficient $\Theta_{x,t}$ depends on parameters and taxes and the exponent σ_x on parameters only.

Proposition 1. *In economic equilibrium we have the following:*

i. Aggregate labor is decreasing in τ_t, ϵ and γ_0 and increasing in τ_{t+1}, θ. Aggregate savings are decreasing in τ_t, τ_{t+1}, and θ and are increasing in ϵ and γ_0.

ii. Labor income inequality and relative labor supply are independent of pension policy and given by

$$\frac{\eta_i h_{t,i}}{h_t} = \frac{\eta_i^{1+\xi}}{X_i^{\xi} \Gamma_h}, \qquad \Gamma_h \equiv \sum_i \frac{\gamma_i \eta_i^{1+\xi}}{X_i^{\xi}}. \tag{12}$$

iii. Savings of formal households relative to average savings are independent of τ_t and linear in τ_{t+1}:

$$\frac{s_{t,i}}{s_t} = \frac{\eta_i^{1+\xi}}{X_i^{\xi}} \frac{1}{\Gamma_h} + \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}}{1+\gamma_0 \epsilon} \frac{1-\theta}{1+\beta} \left(\frac{\eta_i^{1+\xi}}{X_i^{\xi} \Gamma_h} - 1 \right). \tag{13}$$

iv. When $\tau_{t+1} = 0$ or with fully Bismarckian pensions ($\theta = 1$), the heterogeneity in savings exactly matches labor income inequality. An increase in τ_{t+1} or a decrease in θ amplifies the spread in private savings.

v. Informal savings increase in τ_t and θ and decrease in τ_{t+1} and ϵ; current taxes also raise their ratio to formal savings.

Part i of proposition 1 states that current taxes reduce labor supply and savings, while future taxes increase labor supply, since higher expected benefits make work more attractive, and reduce the need to save. A higher θ increases labor supply for the same reason; with GHH preferences this raises the marginal utility of $\tilde{c}_{1,t}$ and so, through the Euler equation, lowers aggregate savings. A higher ϵ shifts future benefits toward informal households and lowers the benefit formal households expect, so that they save more and work less, and a larger share of informal households has the same effects.

Parts ii and iii reduce the state space of the politico-economic equilibrium, as the next subsection shows. Part iv implies that higher future taxes or a lower θ, when pensions are redistributive, widen the dispersion of savings and thus of private wealth. Part v works through the universal pension, since informal earnings do not respond to pension policy. The pension's present value to an informal household is proportional to formal savings, with a factor that rises with future taxes and ϵ: current taxes and a higher θ, which lower formal savings, therefore raise informal savings, while future taxes and a higher ϵ raise the present value of the pension and lower them.

3.2 Politico-Economic Equilibrium

In a Markov equilibrium the distributions of past savings and past labor supply are in principle state variables of the political choice of the tax rate. Part ii of proposition 1 removes the distribution of labor supply, since taxes do not move relative labor supply.[9] If informal households save, their past savings are a further state, and one that both current and future taxes move. For the analytical results we therefore take informal households to be hand-to-mouth: they do not save, so that their Euler condition (6d) does not apply, and when old they receive an endowment tied to the informal wage, which can be read as work in old age at a lower productivity.[10]

With log-GHH preferences and hand-to-mouth informal households, the policymaker maximizes

$$\begin{aligned}\mathcal{W}_t(z_t, \tau_t; \tau_t^{+1}(z_{t+1})) = \sum_i \gamma_i \mu_i \left[\omega \ln\left(c_{2,t}^i\right) + \nu_t \left(\ln\left(\tilde{c}_{1,t}^i\right) + \beta \ln\left(c_{2,t+1}^i\right)\right)\right] \\ + \gamma_0 \mu_0 \left[\omega \ln\left(\tilde{c}_{2,t}^0\right) + \nu_t \left(\ln\left(\tilde{c}_{1,t}^0\right) + \beta \ln\left(c_{2,t+1}^0\right)\right)\right] \qquad (14)\end{aligned}$$

Proposition 2. *In politico-economic equilibrium with hand-to-mouth informal households, all consumption functions are log-separable in the endogenous state variables, mak-*

[9] This property of GHH preferences also holds under CRRA preferences (appendix A.1).

[10] We model old-age income as an endowment of goods to keep the model as symmetric with the general case as possible.

ing policy only a function of parameters, including system characteristics and demographics. Politico-economic equilibrium taxes are increasing in ω and decreasing in ν_t, and, with equal propensities to vote, they are increasing in income inequality and decreasing in θ. The effect of ϵ is ambiguous and depends on the pension's share of consumption of informal and formal retirees.

The economic equilibrium is log-separable in the aggregate states (appendix A.2), and the one remaining vector of states, the distribution of formal households' savings $s_{t-1,i}/s_{t-1}$, is a function of parameters and the current tax alone; by the same token, the distribution carried into the next period does not depend on the current choice (appendix B.1). Despite two dimensions of pension design and heterogeneous households, the politico-economic equilibrium therefore shares the property of Gonzalez-Eiras and Niepelt (2008), Gonzalez-Eiras and Niepelt (2012), Song (2011), and Uchida and Ono (2024), among others, that policy is a function of parameters and demographics only, although the equilibrium conditions are more complex and in most cases can be solved only numerically.

Notwithstanding the orthogonality of current policy with state variables, the trade-offs underlying the equilibrium tax rates are dynamic in nature as they relate contemporaneous policy decisions to capital accumulation and thus future wages and tax revenue. The tractability of the model comes from specifying functional forms that render the factor price elasticities and the derivatives of the indirect utility functions orthogonal to the capital stock and relative labor supply and relative savings decisions independent of current taxes.

The comparative statics in ω and ν_t mirror those in Gonzalez-Eiras and Niepelt (2008) and Song (2011), and the latter also shows that the equilibrium tax rate increases with income inequality. A distinct contribution to the politico-economic literature on social security is the prediction that a more Bismarckian benefit formula weakens political support for taxation. The reason for this is that a higher θ shifts benefits from low-income to high-income retirees, and the former, with their higher marginal utility, weigh more in the political objective, so that retirees' support for taxes declines in the aggregate. A higher ϵ shifts benefits from formal to informal retirees. With log utility a retiree's support for the tax is the pension's share of their consumption, so the shift raises the support of informal retirees and lowers that of formal ones (appendix B.1 gives the exact condition for the sign). Taxes rise when informal retirees are poorer than formal ones and the pension is a comparable share of their consumption, as in Argentina (section 5), because their smaller pension then rises in a larger proportion than formal pensions fall.

When informal households save or preferences are CRRA, the tax rate depends on the state and the equilibrium has to be solved numerically; section 4 describes how.

4 Numerical Methods

This section describes how the politico-economic equilibria of sections 5–7 are computed, with the algorithms and the code in the technical documentation (Berg and Gonzalez-Eiras 2026). The quantitative politico-economic literature computes a time-invariant policy function (Krusell and Ríos-Rull 1999; Forni 2005; Song 2011) or compares steady states across demographic scenarios (Galasso 1999). Along a demographic transition, however, the Markov policy function is date-specific, since the policymaker at t faces its own mix of cohorts, ν_t, and a continuation policy set further along the transition. The distinction is immaterial in the equilibrium of proposition 2, which carries no state, but not under CRRA preferences or with informal savers, where the continuation reaches the tax through the state, or in the design choice, where a design chosen at t is weighed against the electorate of $t+1$.[11] We therefore compute date-specific policy functions by backward induction from a date T (appendix D), the end of a finite horizon or the start of a steady state in an infinite one. For a large T the two agree away from T. We report the finite-horizon solution, which approximates the Markov equilibrium in the limit of finite-horizon economies, the class in which Gonzalez-Eiras and Niepelt (2008) establish uniqueness.

In the general case with CRRA preferences, the distribution of labor supply remains independent of policy (part ii of proposition 1) and the distribution of past savings is pinned down by the current tax together with past aggregate savings (appendix A.1). The distribution of past savings matters for the political problem, but since it depends only on the current tax and past aggregate savings we can eliminate it from the state space, keeping only the level of past savings s_{t-1}: we work with the first order condition in τ_t derived with the distribution of past savings held fixed, and then replace it with the one consistent distribution for each candidate tax and for each state s_{t-1}. Informal savers add the ratio $s_{t-1,0}/s_{t-1}$ of informal to formal savings to the state space, which also depends on τ_{t-1} (appendix B.2).

When the policy function features one or more states, the current tax moves the states and affects future policy choices. We measure the marginal effect by numerical differentiation of the $t+1$ solution grid, which the backward induction has already computed. A first order condition can have several roots, some of them minima, and the optimum can lie at a corner where it has none, so we evaluate the condition on a grid of taxes and test the corners and all its roots. A candidate passes if it maximizes the political objective with its own consistent distribution held fixed.[12]

[11] In our applications, the distinction has a modest effect on results: a time-invariant policy function, solved at each date's own ν_t and applied along the projected path, misses the tax rate by roughly 0.2 p.p., and by up to 0.5 p.p. in the Argentine calibration at the lowest intertemporal elasticity, and misplaces the design by up to 0.059.

[12] Several passing candidates would be ranked by payoff, an assumption about equilibrium selection

Each reported path starts from the steady state under the first period's policy. Parameters whose targets are equilibrium outcomes, such as the discount factor and the political weight of the old, are calibrated by re-solving the equilibrium path at each trial value until the targets of sections 5 and 6 hold in the calibration year.

Under log preferences the choice of the design in section 7 adds no state either. The design chosen at t applies from $t+1$ and depends neither on the design chosen in the previous period, which sets current benefits, nor on aggregate savings (proposition 4 in appendix C.4), and the tax at each candidate design is found as above, without a state. Each date's choice is therefore a search over a grid of designs. Under CRRA preferences neither property survives: the equilibrium is a backward recursion over policy functions of the state (s_{t-1}, θ_t).

5 Pension System Reform in Argentina

Argentina is a middle-income country with high labor informality. In 1994 the social security system was reformed tying full benefits to 30 years of contributions, which left about a third of the elderly without a pension by 2004.[13] From 2005 a series of tax amnesties let workers with shorter contribution records draw a pension. They added 2.7 million beneficiaries by 2011 (Rofman and Apella 2015), raising the number of pensioners from about 3 to 5 million (figure 5.1) and the coverage of the elderly from 68% to over 90% (Rofman and Oliveri 2012). Thus, in a few years, Argentina experienced a significant increase in non-contributive pensions: Between 2005 and 2010 social security spending increased by 1.9% of GDP with most of this increase, 1.8% of GDP, due to the tax amnesties (Cetrángolo and Grushka 2020).

The following outlines the calibration procedure and presents the main results from the pension reform. Appendix E keeps calibration details and the online appendix provides supplementary evidence from the reform.[14]

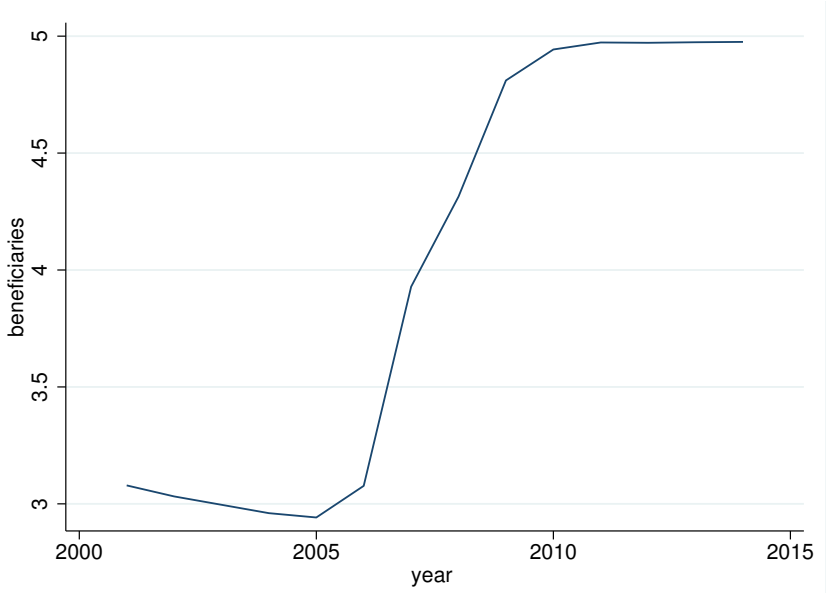
Calibration. We calibrate the model with informal savers to Argentina before the reforms and read the tax amnesties as an increase in ϵ. Informal households work when young and save for retirement, facing the same rate of return on savings as formal households. A model period is thirty years, the calibration year is 2010, and every household has the same propensity to vote. Table 1 reports the result. The pre-reform $\epsilon = 0.21$ is the basic pension of retirees without a full contribution record, 70% of the benefit of the

that none of our reported equilibria requires: exactly one passes at every state and date of every solve that records the count (online appendix).

[13] The reform also let workers opt for a fully funded system, whose funds were nationalized in December 2008.

[14] Appendix E holds the data, the identifying formulas and the sources of the calibration. The online appendix has additional evidence including how the design (ϵ, θ) shapes equilibrium outcomes and how different calibrations affect results.

Figure 5.1: Pension coverage in Argentina



Note: Beneficiaries of the national pension system (SIPA) and of non-contributive old-age pensions, annual averages in millions. Source: Cetrángolo and Grushka (2020, table 6), from the Boletín Estadístico de la Seguridad Social.

least productive formal workers, in units of the average formal benefit and discounted because it starts five to ten years after the formal retirement age, and $\gamma_0 = 0.47$, informal households per formal household, puts informal households at 32% of all, the share of the population over 65 without a pension before the reform (Rofman and Oliveri 2012). The Bismarckian index $\theta = 0.84$ matches the observed ratio of replacement rates at mean and half-mean income. The reform raises ϵ to the universal level $\epsilon^U = 1 - \theta + \theta\eta_1 h_{t-1,1}/h_{t-1}$ of section 2, at which informal retirees receive the benefit of the least productive formal ones. We impose a formal-sector capital share of $\alpha = 0.35$, between the values implied by the survey's informal earnings and by shadow-economy estimates of informal output, a Frisch elasticity of $\xi = 0.30$, and a ratio of workers to retirees ν_t from thirty-year gross population growth in census data and projections, as in Gonzalez-Eiras and Niepelt (2008).

The remaining parameters are identified jointly with the equilibrium path. The formal productivities η_i match the income distribution across the four formal quartiles of the survey, with a taste for leisure X common to them whose level is pinned by the observed formal workweek of 42.5 hours.[15] Relative hours across quartiles are then a prediction, which reproduces the survey's ranking and overstates the hours of the top quartile, 1.18 against 1.10 times the formal average. Four parameters are solved against four targets in

[15]Identifying one X_i per quartile instead takes the survey's relative hours as data and leaves the level of hours unidentified (appendix E.5). Since the productivity distribution reaches the aggregates only through a sum that both calibrations normalize to one, only η_i and X_i differ, and every result of this section is the same under both.

2010: β against a capital-output ratio of 3.23, ω against an equivalent social security tax rate of 10.9%, which is pension spending of 7.1% of GDP before the 2001 crisis divided by the labor share of formal output, and η_0 and X_0 against the informal households' income and hours relative to the formal average.

Table 1: Calibration, Argentina

Parameter	**Value**	**Target**
ϵ (*pre*-reform)	0.21	Basic pension, 70% of the minimum, discounted
θ	0.84	Replacement rate dispersion
α	0.35	Factor income shares
ν_t	$[1.65, 0.96]$	30-year gross population growth rates
ξ	0.30	Elasticity of labor supply
β	0.65	Capital–output ratio of 3.23
X	1.94	Average formal workweek of 42.5 hours
η_i	$[0.64, 0.90, 1.14, 1.87]$	Income distribution
γ_0	0.47	Elderly without a pension, 32% of households
ω	1.45	Social security tax of 10.9%
η_0	0.344	Informal relative income, eq. (E.1)
X_0	0.804	Informal relative working hours

Note: Our default specification relies on $\rho = 1$.

The reform. Table 2 reports the effect in 2010 of an unanticipated and permanent rise of ϵ to ϵ^U under log-GHH preferences, first with the tax rate held at its pre-reform path and then chosen politically. We find that taxes increase, which drives down savings and labor supply. With taxes held fixed, formal households save more and work marginally less, since the reform lowers the benefit they expect (proposition 1); informal households, who now hold a claim on the pension system, save less.

In 2010 the model predicts that the social security tax rate rises by 2.1 p.p., which at the labor share of formal output is pension spending of 1.4% of GDP, about three quarters of the 1.8% of GDP that the amnesties added. The reason for this is that the reform moves benefits toward informal retirees, who are poorer than formal ones and value benefits more at the margin.[16] The savings rate falls by 0.4 p.p. of GDP and the average workweek by a quarter of an hour, the fall in savings in line with the estimate of Gonzalez-Rozada and Ruffo (2016) for the amnesty. The tax response builds over time, to 3.5 p.p. above the pre-reform path by 2040, because informal households, now covered, save less for retirement and as retirees demand more taxation.

[16]The tax also rises when informal households are hand-to-mouth, consistent with condition (B.5) in appendix B.1.

Table 2: Pension system reform, year 2010

Scenario	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	10.92%	14.40%	42.54
Economic Equilibrium	10.92%	14.73%	42.46
Full effect	13.00%	13.97%	42.29

Note: The reform permanently raises ϵ to $1-\theta+\theta\eta_1 h_1/h$ in the calibration year, unanticipated. The "Economic Equilibrium" scenario solves for the new economic equilibrium path with taxes as in the Baseline scenario, but with the new permanent ϵ; the "Full effect" scenario lets taxes be determined by the politico-economic equilibrium. $\rho = 1$. The savings rate is savings relative to GDP.

The effect of the intertemporal elasticity of substitution. Since the environment has no uncertainty, we read the parameter ρ of CRRA preferences as the intertemporal elasticity of substitution (IES). The evidence on its size is inconclusive,[17] and we keep logarithmic utility as the benchmark. Figure 5.2 repeats the reform with the pre-reform economy calibrated at each ρ and traces the effects in 2010 and 2040 for taxes, savings rates, average workweek, and informal savings.

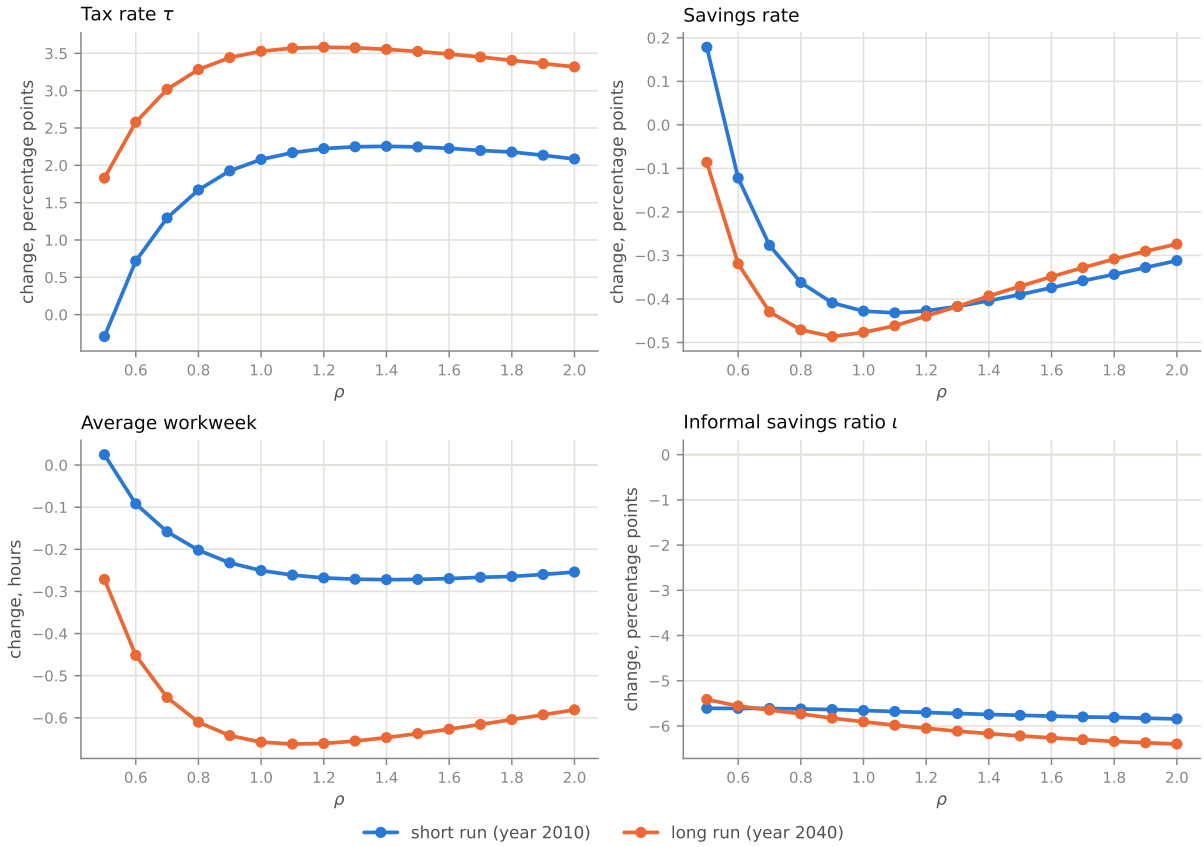
The effects of the pension reform are nonlinear in ρ, in particular for values lower than one. When $\rho < 1$ the effects of the pension reform are dampened across all metrics, and at the lowest values of the IES the tax rate even falls slightly on impact. The reason for this is that with a low IES the young resist taxation more strongly, so the tax rate responds less to a change in benefits. For an IES between 1 and 2 the results stay close to the log benchmark: the tax rises by 2.1 to 2.3 p.p. in 2010, the workweek shortens by roughly 0.25 hours in 2010, and the fall in the savings rate shrinks from 0.4 to 0.3 p.p. of GDP. The informal savings ratio falls by 5.7 to 5.8 p.p. throughout this range. Results are similar in 2040, albeit with tax and workweek effects amplified. Since Gonzalez-Rozada and Ruffo (2016) find that the amnesty reduced savings, we regard $\rho \in [1, 2]$ as the more plausible range.

6 Social Security in Rich OECD Countries

In order to study the determinants of pensions, we now calibrate the model to three rich OECD economies that differ markedly in pension spending, demography, income inequality, voting patterns and pension design: the U.S., France and the UK. In these economies nearly every worker qualifies for a contributive pension, so we set informality aside and

[17] Estimates from aggregate consumption data are low, which Guvenen (2006) reconciles with higher values from output and wealth data through limited asset market participation, and asset-pricing models with Epstein–Zin preferences and long-run or disaster risk require an IES above one (Bansal and Yaron 2004; Barro 2009).

Figure 5.2: Short- and long-run effects of the pension reform as a function of ρ



Note: Each panel is the change against the pre-reform path of the same ρ. The savings rate is savings relative to GDP, and the informal savings ratio ι is the savings of an informal household relative to the average savings of formal households.

the design reduces to one dimension, the extent to which benefits are earnings-related.[18] To explore what characteristics shape pensions, savings rates, and labor supply, we simulate counterfactual economies that are calibrated to the U.S. with one characteristic changed at a time.

As for the Argentina case, the following outlines the main results. Appendix F and the online appendix contain more details on the calibration and supplementary evidence.[19]

Calibration. We calibrate the three economies to 2020 in the same way as Argentina in section 5, with a few exceptions. First, with informality set aside, the population is split into three formal income groups. The first two groups are cut, as for Argentina, to have mean incomes of half the mean and the mean, the incomes at which the OECD's replacement rates identify θ. France's system is fully Bismarckian, $\theta = 1$ at any cuts, so

[18] Whitman et al. (2011) estimate that about 4% of individuals aged 62–84 in the U.S. in 2010 would never collect Social Security, most of them for lack of the 40 quarters of covered earnings needed to qualify.

[19] The online appendix holds the calibrations across the intertemporal elasticity of substitution and under the alternative treatment of the taste for leisure, and the counterfactuals of this section with the UK as the host economy and across the Frisch elasticity, each figure with the data behind it.

we group French households at the U.S. income percentiles instead, which puts the two countries' household types in one-to-one correspondence for the counterfactuals below. Second, the discount factor β matches the U.S. thirty-year interest rate rather than a capital–output ratio and is imposed on France and the UK. Third, where the Argentine calibration gives every household the same propensity to vote, the political objective here weights each income group by its observed propensity. In the U.S. the poor vote much less than the rich, while in France the propensity to vote is nearly flat across income groups. Fourth, since the model is one of intergenerational transfers, we add Medicare to U.S. pensions, in spending and in the calibration of θ, where as a benefit unrelated to income it makes transfers to the old more Beveridgean than pensions alone.

Table 3 reports the parameters. Population growth is highest in the U.S. and lowest in France, $\nu_{2020} = 1.34$ against 0.97, and the ratio of high to low productivity is about a third higher in the U.S. than in France, 3.08 against 2.29. The average taste for leisure is roughly twice the U.S. level in the UK and France. The calibrated political weight of the old is nearly the same in the U.S. and France, 1.45 and 1.42, and lower in the UK, 1.16.

Table 3: Calibration, the U.S., the UK and France

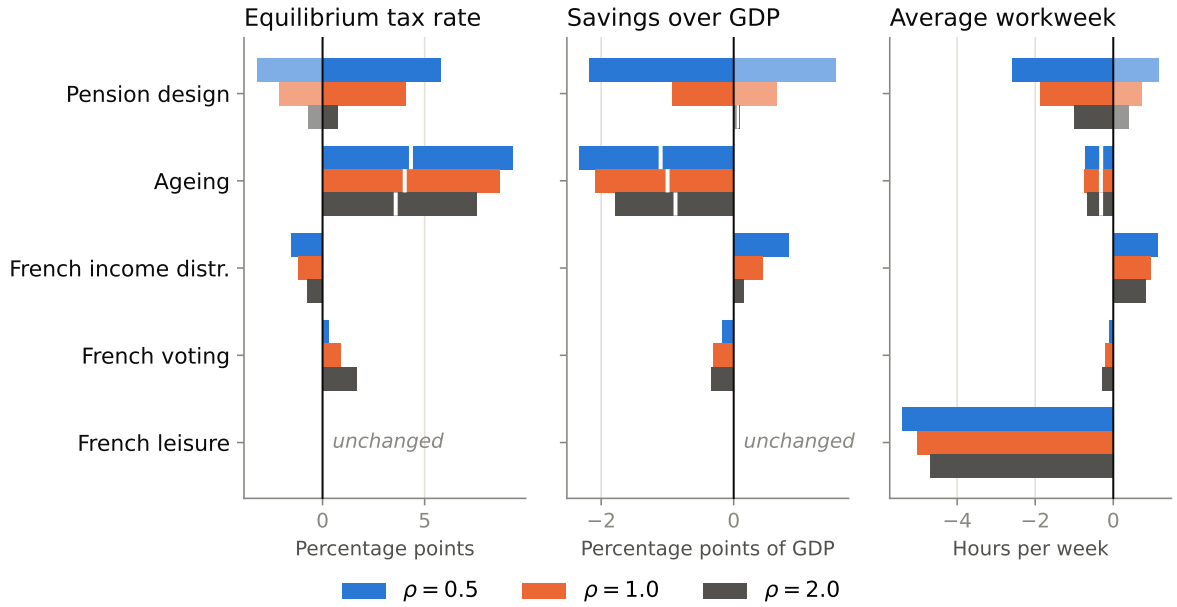
Parameter	**U.S.**	**UK**	**France**	**Identified by**
θ	0.74	0.56	1.00	Replacement rate dispersion
ω	1.45	1.16	1.42	$\tau^{US} = 14.4\%$, $\tau^{UK} = 11.9\%$, $\tau^{FR} = 21.3\%$
X	4.03	8.26	7.27	Average workweek
ν_{2020}	1.34	1.17	0.97	30-year gross population growth
η_H/η_L	3.08	3.06	2.29	Relative productivity, high to low income
β		0.76		30-year interest rate (U.S.)
ρ		1		Log-GHH preferences
ξ		0.30		Frisch elasticity of labor supply
α		0.30		Capital income share

Note: X is the population-weighted mean of X_i; its level is the hours unit, pinned for France and the UK by targeting average hours relative to the U.S. rather than in levels. The lower panel is common to the three countries; β is calibrated for the U.S. and imposed on the other two.

Determinants of pensions, savings, and labor supply. Figure 6.1 summarizes the counterfactuals. Each row changes one characteristic of the U.S. calibration: its pension design, population growth, income distribution, leisure preferences or voting patterns. Each panel reports the effect on the equilibrium tax rate, the savings rate and

the workweek in 2020, at the three values of the intertemporal elasticity ρ considered below. The counterfactuals are guided by France and the UK, particularly France, whose households can be grouped as in the U.S. and whose calibrated ω is close to the U.S. value. Each is a separate equilibrium path on which the changed characteristic holds throughout, so a row describes an economy that has always had that characteristic rather than the U.S. surprised in 2020.

Figure 6.1: The effect of various counterfactuals in the U.S.



Note: The online appendix repeats the figure with the UK as the host economy and gives the data behind both. In the pension-design row each bar shows both corners, measured from the baseline: $\theta = 0$ in full colour and $\theta = 1$ in the light tone. In the ageing row the bar is acute ageing and the white line across it mild ageing. The savings rate is savings relative to GDP.

The tax panel identifies the pension design and ageing as the main determinants of pension spending. Going from a fully Bismarckian system to a fully Beveridgean one raises the equilibrium tax rate by 6.2 p.p. at $\rho = 1$ and lowers the savings rate by 1.6 p.p. of GDP. The reason for this is that of proposition 2: flatter benefits direct more of the revenue to poorer retirees, whose higher marginal utility gives them more weight in the political objective, so retirees support a larger system. Holding taxes fixed, an increase in θ would lower savings, but the endogenous fall in taxes more than offsets this and the net effect is positive. For labor supply, the economic-equilibrium and political effects of an increase in θ are aligned, and the tax response contributes as much as the direct effect or more.

Population ageing is the determinant that previous studies of the politico-economic equilibrium of pension spending have highlighted: Gonzalez-Eiras and Niepelt (2008) predict an increase in U.S. spending of around 3.6 p.p. towards the middle of this century because of the slowdown in population growth. In the data, including projections, ageing

is more acute in France than in the UK, and much more than in the U.S. We consider two scenarios, which roughly correspond to the data for the two European countries. In the first, *mild ageing*, we halve current and projected U.S. population growth; in the second, *acute ageing*, the population is constant, $\nu_t = 1$ for all t. Acute ageing raises the tax rate by 8.7 p.p. and mild ageing by 4.0 p.p.; the mild scenario is close to the demographic projection of Gonzalez-Eiras and Niepelt (2008).[20] When acute ageing drives up pension contributions significantly, this crowds out private savings by 2.1 p.p. of GDP at $\rho = 1$. Working hours fall by roughly 0.7 hours a week: the higher tax rate lowers the effective wage rate net of taxes, but this is partly offset by a capital deepening effect from ageing on wages. Thus, the net result is a modest drop in hours worked.

The characteristics that set France's households apart move the tax rate much less. At $\rho = 1$, France's flatter income distribution, imposed with θ held at its U.S. value, lowers it by 1.2 p.p.,[21] and its flatter voting profile, under which the poor vote about as often as the rich, raises it by 0.9 p.p. Rescaling the taste for leisure to France's level shortens the workweek by more than the whole gap to France's and leaves every rate at the baseline, since a proportional change in X_i scales hours, output and savings together. With the UK as the host economy the effects are smaller still, and France's income distribution raises the UK's tax rate, since at the UK's income cuts France is the more unequal of the two countries (appendix F.4).

The workweek panel ranks the determinants differently. France's leisure preferences move hours most and leave every rate unchanged. Among the characteristics that move the tax rate, the pension design moves hours most, by about two hours a week at the Beveridgean corner at $\rho = 1$, while ageing, the largest effect on taxes, moves them less than France's income distribution does at every ρ. The determinants of pension spending are thus not the determinants of hours worked.

The effect of the intertemporal elasticity of substitution. In general, our results above are robust across different ρ and the overall conclusion about the determinants of pension spending, savings, and labor supply holds. Effects are generally slightly amplified for lower values of ρ and dampened for higher values of ρ; the one exception where this pattern is reversed is the French voting profile. The IES plays a key role in the effect of the pension design: going from $\theta = 1$ to $\theta = 0$ raises the tax rate by 9.0 p.p. at $\rho = 0.5$ and by only 1.5 p.p. at $\rho = 2$. The reason for this is that θ moves the tax rate through the difference in marginal utility between poor and rich retirees (proposition 2), and a higher IES makes marginal utility fall less steeply with consumption, so that this

[20] In Gonzalez-Eiras and Niepelt (2008), $\nu_{2000} = 1.38$ and the projected $\nu_{2080} = 1.23$, a fall of about 0.15, similar to the fall of 0.17 in 2020 in the mild ageing scenario.

[21] We read θ off the ratio of replacement rates at mean and half-mean income, so under France's flatter income distribution the same observed ratio would imply a considerably less Bismarckian system. Re-estimating θ would bundle a change in the design into a counterfactual about inequality.

difference counts for less; the savings rate and the workweek follow.

The calibrated parameters absorb institutions that the model does not contain. The political weight of the old, ω, stands for the way legislation is introduced, debated and passed, and for constitutional arrangements such as whether the political system is presidential or parliamentary. The taste for leisure, X, stands for preferences and also for institutions that shape labor supply, such as unionization, collective bargaining, labor-market regulation and statutory holidays, so the large differences in X across the three countries include these institutional differences.

We now return to figure 1.1, in which pension spending and the Bismarckian index both rise as population growth falls, and neither is related to income inequality. With the design given, the model accounts for the pattern in pension spending: ageing is the main determinant of the equilibrium tax rate, and the income distribution moves it several times less.[22] The relation between the design and population growth suggests that countries respond to ageing by tying benefits more closely to earnings, which a model that holds θ fixed cannot address. Section 7 takes it up by letting the electorate choose θ as well as τ.

7 Endogenous Pension Design

Up to this point the electorate chooses the size of the pension system while its design is calibrated. In this section it chooses the Bismarckian index θ as well as the tax rate, with the same political objective $\mathcal{W}_t$, in the economy of section 6 without informal households. Appendix C holds the derivations, appendix G further results, and the technical documentation the algorithms (Berg and Gonzalez-Eiras 2026).

7.1 The deadweight cost of redistribution

Left to choose the design in the model as it stands, the electorate chooses flat benefits. For a given revenue, flattening benefits shifts income from retirees who earned more than the average when young to those who earned less, and the political objective, which weighs retirees by their marginal utility, counts the gains of the poorer for more than the losses of the richer. Proposition 3 in appendix C.1 makes the argument exact: as long as the rich are not sufficiently more likely to vote than the poor, an electorate that sets the design and the tax at the same time picks the Beveridgean corner, $\theta_t = 0$, whatever the tax rate, the demography and the income distribution. In the U.S. the rich do vote more often, which could in principle reverse the result, but at the U.S. calibration each timing of the choice that we examine still ends at a corner; this also holds if we let the

[22] On the disposable-income Gini of figure 1.1, the U.S., at 0.42, is the third most unequal of the 29 countries, after Mexico and Turkey, and France, at 0.32, is the median.

electorate choose pension designs in advance or permanently (appendix C.1).

The model does contain a cost of flat benefits. A flat benefit does not reward an extra hour of work, and at the Beveridgean corner the workweek is about two hours shorter than at the observed U.S. design (section 6). When the design is chosen ahead of the benefits it governs, the electorate weighs this cost, and retirees, whose benefits are levied on current labor, favor an earnings-related design because of it. At U.S. inequality the cost is too weak to hold the electorate back. What the model misses is a mechanism through which flat benefits cost more than earnings-related ones on margins other than hours.

The empirical literature points to such margins. Disney (2004) splits public pension contributions in OECD countries into a part matched by benefit entitlements and a part that is not, and finds that the second lowers the labor force participation of women while the first raises it. Kumler et al. (2020) find that after Mexico's 1997 pension reform tied benefits more closely to the wages reported to the social security agency, the under-reporting of wages fell among the younger workers the reform affected. Benefits that are loosely tied to contributions thus erode the base on which contributions are levied, through participation in the formal labor market and through the reporting of earnings, two margins the model does not contain.

Summers (1989) explains why the link matters. The part of a payroll contribution that buys the contributor a benefit is closer to deferred pay than to a tax, while the part that buys nothing distorts like a tax. Specifically, with $y_i \equiv \eta_i h_{t-1,i}/h_{t-1}$ the earnings of type i relative to the mean, constant over time by part ii of proposition 1, a retiree of type i receives $\nu_t w_t h_t \tau_t \left[\theta_t y_i + 1 - \theta_t\right]$ on past earnings $w_{t-1} y_i h_{t-1}$, that is

$$\Pi_t \, \tau_t \left[\theta_t + \frac{1-\theta_t}{y_i}\right] \text{ per unit of past earnings,} \qquad \Pi_t \equiv \frac{\nu_t w_t h_t}{w_{t-1} h_{t-1}},$$

where Π_t is the pay-as-you-go return factor: a purely earnings-related system paying the same revenue returns $\Pi_t \tau_t$ per unit of earnings to every type. The flat component moves type i away from that benchmark by

$$t_i \equiv \Pi_t \, \tau_t \left(1-\theta_t\right) \left(1 - \frac{1}{y_i}\right) \tag{15}$$

per unit of earnings.

We do not model these extensive participation and reporting margins explicitly. Instead, we borrow a principle from the literature on the deadweight cost of taxation, that the loss a tax causes rises with the square of its rate (Harberger 1964). Applied to the implicit tax of each type, we model this by taking a loss out of the revenue that reaches

retirees, so that the budget becomes (see appendix C.2 for the derivation)

$$\nu_t w_t \tau_t h_t \, f(\theta_t, \tau_t) = \sum_i \gamma_i b_t^i, \tag{16}$$

$$f(\theta_t, \tau_t) = \exp\!\left(-\tfrac{1}{2}\,\lambda\,\tau_t\,\tilde{V}\,(1-\theta_t)^2\right), \qquad \tilde{V} \equiv \sum_i \gamma_i \frac{(y_i-1)^2}{y_i},$$

where $\lambda \geq 0$ is a cost parameter to be calibrated, $1-f$ the share of revenue lost and $\tilde{V}$ the dispersion of relative incomes.[23] We see this form as a reasonable place to start rather than as an exact account of participation and reporting: it has the properties that the principles above suggest, and it leaves the size of the loss to the calibration. The loss vanishes when the system redistributes nothing, at $\theta_t = 1$ or when all incomes are equal, since flat and earnings-related benefits then pay everyone the same. Its share of revenue grows with the tax rate, so each unit of redistribution costs more in a larger system. And since the loss is quadratic in the flat share, the first unit of redistribution costs nothing at the margin, and the cost alone cannot take an electorate to a fully Bismarckian system.

The electorate chooses the design one period ahead: at t it sets θ_{t+1}, which governs the benefits of the current young. A design chosen together with the tax would only re-split benefits that current retirees have already earned, and so would move no decision, while a design chosen once and for all would require a commitment that the Markov equilibria of section 2 rule out. Reforms of benefit formulas, moreover, usually apply to benefits not yet in payment. Formally, the cost only replaces θ_t and $1-\theta_t$ by $f\theta_t$ and $f(1-\theta_t)$ in the equilibrium of the previous sections (appendix C.3). Since it scales both components of the benefit alike, it cancels from the ratio of replacement rates that identifies the design in section 6, and the observed U.S. design is still $\theta^* = 0.738$. In the tax condition it adds one factor: a larger system loses a larger share of its revenue, so a retiree gains less from a higher tax than the revenue it raises. And a more earnings-related design now also enlarges the benefits the young will receive, because less of the revenue that pays them is lost; this gain is what can stop the electorate short of the Beveridgean corner. Under log preferences, the design chosen at t depends neither on the design inherited nor on savings (proposition 4 in appendix C.4), so each date's choice is a search over designs without a state (section 4).

7.2 Calibration and the determinants of the design

Calibration. We calibrate the cost parameter so that the electorate re-elects the observed design. Concretely, the target is θ_{2020}, set by the electorate of 1990 on an equilibrium path along which every design from the first period on is chosen, and β and ω are

[23] The exponential keeps f between zero and one for any size of the system and gives back the model without the cost at $\lambda = 0$; to first order, $1-f$ is the quadratic loss itself.

recalibrated at every trial value of λ, so that the targets of section 6 still hold with the cost in place. Only the design of 2020 is targeted; every other design along the path is an outcome of the model.

Table 4 reports the calibration at each intertemporal elasticity. The table shows that at the calibrated design, the loss amounts to a relatively small share of revenue, no more than 4% ($1 - f(\theta^*)$). However, the calibrated cost parameter λ varies by roughly an order of magnitude across the reported range, and so does the share of revenue that a completely flat benefit would lose ($1 - f(0)$). The cost parameter falls as the elasticity rises, because a higher elasticity strengthens the forward-looking stakes in an earnings-related design, and less cost is needed to keep the electorate at the observed level. At every elasticity λ lies above what participation and taxable-income elasticities alone would imply, by more than an order of magnitude at $\rho \leq 1$, so the cost also stands in for administrative, evasion and political costs of redistribution that the model does not contain (appendix C.2). The parameter calibrated on the U.S. accounts for much of the UK's distance from the U.S. but none of France's, whose corner no value of the parameter reproduces (see appendix G.1).

Table 4: The calibrated cost of redistribution for the U.S.

ρ	λ	ω	$1 - f(\theta^*)$	$1 - f(0)$
0.5	18.267	2.50	3.9%	43.7%
1.0	8.643	1.51	1.8%	23.8%
2.0	1.724	1.20	0.4%	5.3%

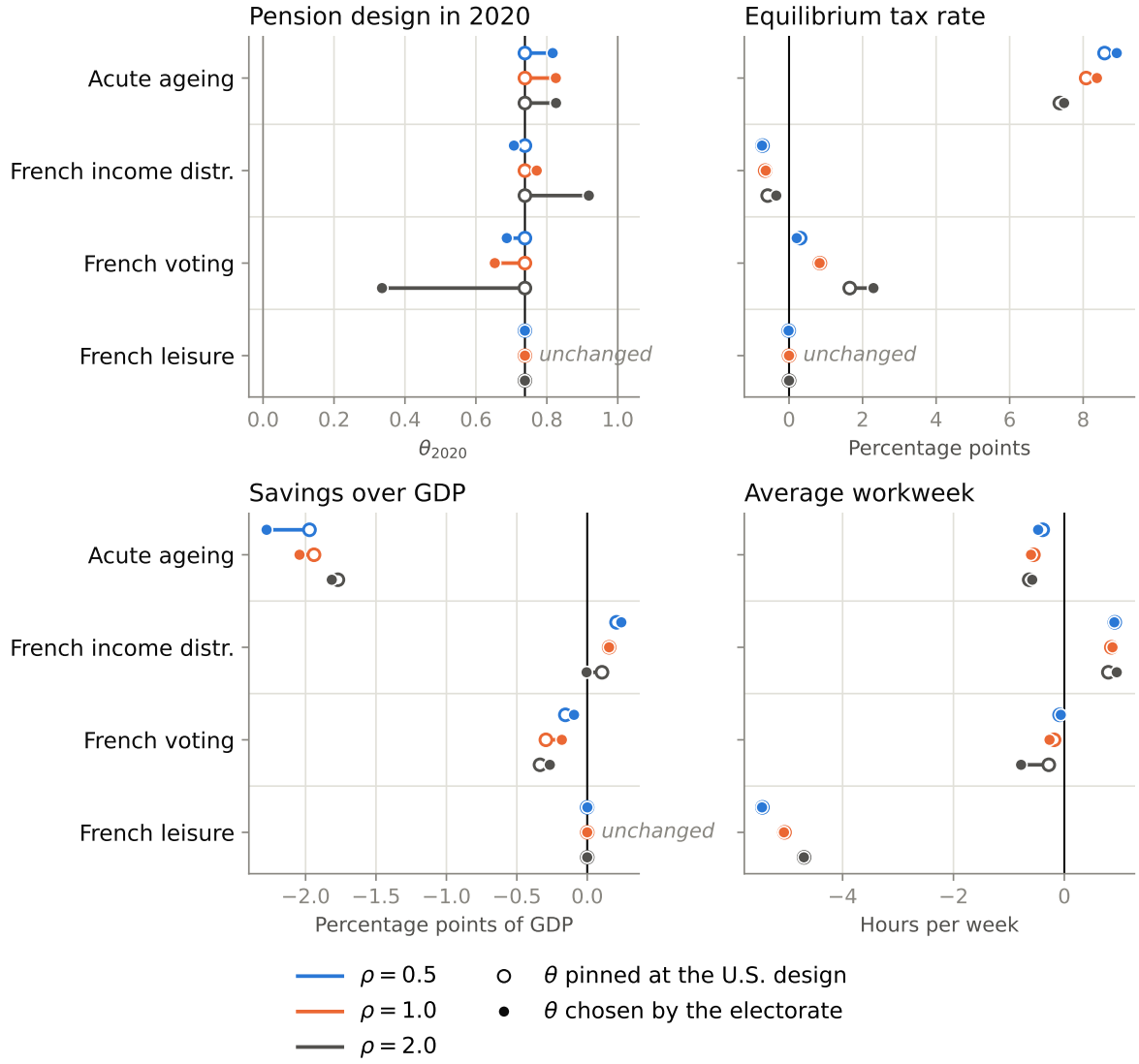
Note: The cost is that of equation (16), with $\theta^* = 0.738$ and $\tilde{V} = 0.436$ at every ρ. $1 - f(\theta^*)$ and $1 - f(0)$ are the shares of revenue lost in 2020 at the observed design and at a flat benefit, both at the 2020 tax rate. β and ω are recalibrated with λ.

Determinants of the pension design. Figure 7.1 takes the counterfactuals of section 6 to the economy with the endogenous pension design: acute ageing and France's income distribution, voting patterns and taste for leisure, each imposed on the U.S. calibration over the whole horizon and read in 2020. Every counterfactual is read twice, with the design pinned at θ^* and with the design chosen. Because the cost is present in both, the pinned reading departs from figure 6.1, but results do not move materially.[24]

First, the results from section 6 about the determinants of the size of the pension system, savings and labor supply remain mostly unchanged. The biggest change is that ageing with an endogenous design now crowds out private savings more, in particular for low values of ρ. Second, the panel on the pension design shows that ageing consistently ties benefits more closely to earnings, that the French voting profile moves the design

[24]For instance, with the design pinned at $\rho = 1$, acute ageing raises the tax rate by 8.1 p.p. rather than 8.7 p.p., since the cost makes a larger system worth less to retirees at the margin.

Figure 7.1: The effect of various counterfactuals with the design chosen



Note: The model is recalibrated for each ρ under the cost equation (16).

towards more flat benefits, and that a more equal income distribution moves the design little except at $\rho = 2$.

Ageing consistently ties benefits more closely to earnings through two forces. First, ageing gives retirees more weight in the political objective, and retirees prefer a design that raises current labor supply, on which their benefits are levied, which is the more earnings-related one. Second, the higher tax rate associated with ageing raises the loss on every unit of redistribution, since the share of revenue lost grows with the size of the system. Thus, the higher tax increases the incentive to make pension benefits more earnings-related, consistent with the evidence from figure 1.1.[25]

[25]This last channel supplies about half of the response: under a cost attached to the design itself, which does not grow with the size of the system, the design rises only to 0.778 (online appendix). How much the loss grows with ageing rests on the tax rate alone: λ absorbs the pay-as-you-go return factor,

A more equal income distribution moves the design through opposing forces, whose balance depends on the intertemporal elasticity of substitution. On one hand, compressing incomes shrinks the gain that poorer young voters expect from flat benefits in their own retirement, while the stake in an earnings-related design that works through labor supply does not depend on the dispersion of incomes, which pushes the design towards more Bismarckian benefits. On the other hand, compressing incomes makes flat benefits cheaper, since the share of revenue lost scales with the dispersion of relative incomes, which France's distribution halves, from 0.436 to 0.218; this pushes the design towards more Beveridgean benefits. The second force is strongest at low elasticities, where the calibrated cost parameter is largest (table 4), and the first at high elasticities, where the stake through labor supply is strongest. A more equal distribution therefore lowers the design slightly at $\rho = 0.5$, raises it slightly at $\rho = 1$, and raises it by much more at $\rho = 2$.

We return to figure 1.1, in which pension spending and the Bismarckian index both rise as population growth falls and neither is related to income inequality. With the design given, section 6 accounts for the pattern in spending. With the design chosen as well, one political process accounts for both: ageing raises the tax rate and ties benefits more closely to earnings at every intertemporal elasticity, while the income distribution moves the tax rate little and moves the design in a direction that depends on the elasticity, which leaves no systematic relation between inequality and the design.

8 Conclusions

This paper studies how two features of a pension system, whether benefits require a contribution record and how closely they follow past earnings, shape its political support and through it taxes, savings and labor supply. The setting is an overlapping generations model with capital, heterogeneous households, an informal sector and probabilistic voting, first over the size of the system and then over its design as well.

With log-GHH preferences and hand-to-mouth informal households the equilibrium tax is a function of parameters and demographics alone. It rises with the political weight of the old, with ageing and with income inequality, and it falls when benefits are more closely tied to earnings, since earnings-related benefits move revenue from poor retirees to rich ones; universal pensions raise the tax when informal retirees are poorer than formal ones and the pension is a comparable share of their consumption. Calibrated to Argentina, the model accounts for about three quarters of the rise in pension spending that followed the expansion of coverage of 2005 to 2010, with the savings rate lower by 0.4 p.p. of GDP and the workweek by a quarter of an hour. In the calibrations to the U.S., France and the UK, ageing is the main determinant of the tax rate and the

which falls as the population ages (appendix C.2), and a cost that kept the factor would grow less.

earnings link the second; income inequality and voting patterns matter little, and a higher intertemporal elasticity of substitution changes only the size of the earnings-link effect.

When the electorate chooses the earnings link as well, costless redistribution puts the choice at a corner: with propensities to vote that do not rise with income the design chosen with the tax is fully Beveridgean, and at U.S. inequality every timing of the choice gives a corner. An interior design needs flat benefits to be costlier than earnings-related ones, and we read the flat component as an implicit tax whose deadweight loss rises with the square of its rate, so that the share of revenue lost grows with the size of the system. With this cost calibrated on the U.S. design alone, ageing makes the chosen design more Bismarckian at every intertemporal elasticity of substitution, France's flatter voting profile makes it less so, and a more equal income distribution moves it in a direction that depends on that elasticity. The U.S. cost parameter accounts for much of the UK's distance from the U.S. design, and the UK's own parameter is within 17% of the U.S. value at intertemporal elasticities of one or below, though not at two. This is the pattern of the OECD cross-section: spending and the earnings link rise together as population growth falls, and neither follows inequality.

Two limitations point to future work. The model has no risk; in Berg and Gonzalez-Eiras (2023) we study a related environment in which health risk may prevent households from qualifying for a contributive pension, and find that households work and save more in response. And the cost specification cannot deliver France's fully Bismarckian system, because under a cost quadratic in the redistribution performed the first unit of redistribution is free at the margin; a linear term in the cost, or a formality margin, toward which the informal sector of our Argentine model is a first step, would allow it. The same margin bears on the rise of platform work and other arrangements outside contributive coverage in rich economies, which the model can address.

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Appendices

A Economic Equilibrium

The appendix derives results for the economic equilibrium with CRRA-GHH and log-GHH preferences and where informal workers' endowment is tied to wages in the formal sector absent taxes.

A.1 With CRRA-GHH preferences

Household solutions. With CRRA-GHH preferences, the first order conditions in (6) can be written as

$$h_{t,i} = \left(\frac{\eta_i w_t (1-\tau_t) + \frac{\theta \bar{b}_{t+1} \eta_i}{R_{t+1}}}{X_i} \right)^{\xi}$$
$$c^i_{2,t+1} = \tilde{c}^i_{1,t} \left(\beta R_{t+1}\right)^{\rho}.$$

Using the budget constraints and rearranging slightly, we can write individual savings as

$$s_{t,i} = \frac{B_{t+1}}{1+B_{t+1}} \left[w_t \left(1-\tau_t\right) h_{t,i} \eta_i - \frac{X_i \xi}{1+\xi} \left(h_{t,i}\right)^{\frac{1+\xi}{\xi}} \right] - \frac{b^i_{t+1}/R_{t+1}}{1+B_{t+1}},$$

where we have used the shorthand notation $B_{t+1} \equiv (\beta)^{\rho} \left(R_{t+1}\right)^{\rho-1}$.

When informal households save using informal technology, similar conditions apply:[26]

$$h_{t,0} = \left(\frac{\eta_0 w^0_t}{X_0} \right)^{\xi} \tag{A.1}$$
$$c^0_{2,t+1} = \tilde{c}^0_{1,t} \left(\beta R^0_{t+1}\right)^{\rho}. \tag{A.2}$$

As for formal households, we can derive the following for informal savings:

$$s_{t,0} = \frac{B^0_{t+1}}{1+B^0_{t+1}} \left[w^0_t h_{t,0} \eta_0 - \frac{X_0 \xi}{1+\xi} \left(h_{t,0}\right)^{\frac{1+\xi}{\xi}} \right] - \frac{b^0_{t+1}/R^0_{t+1}}{1+B^0_{t+1}},$$

where we have used $B^0_{t+1} \equiv (\beta)^{\rho} \left(R^0_{t+1}\right)^{\rho-1}$. When informal households are hand-to-mouth and receive an endowment when old, the solution for labor supply is given by (A.1).

Savings and labor supply in equilibrium. Next, in economic equilibrium, we impose conditions for factor prices in equations (3) and the specification of pension benefits

[26] Note that given our assumption on universal pensions, informal workers do not perceive a link between their work effort and the benefits they might receive.

including the balanced budget condition in equations (8). We can then rewrite type i's labor supply decision as

$$h_{t,i} = \left(\frac{\eta_i}{X_i}\right)^{\xi} \left(\frac{1}{h_t}\right)^{\xi} \left((1-\alpha)(1-\tau_t)\left(\frac{s_{t-1}}{\nu_t}\right)^{\alpha} h_t^{1-\alpha} + \frac{1-\alpha}{\alpha}\frac{\theta}{1+\gamma_0\epsilon} s_t \tau_{t+1}\right)^{\xi}.$$

The only type-specific term is $(\eta_i/X_i)^{\xi}$; this shows that the ratio $h_{t,i}/h_t$ is only a function of parameters, namely that

$$\frac{h_{t,i}}{h_t} = \frac{(\eta_i/X_i)^{\xi}}{\Gamma_h}, \qquad \Gamma_h \equiv \sum_{i>0} \frac{\gamma_i \eta_i^{1+\xi}}{X_i^{\xi}}. \tag{A.3}$$

Next, we can use the individual labor decision and wage rates to write

$$(1-\alpha)(1-\tau_t) h_t^{1-\alpha} \left(\frac{s_{t-1}}{\nu_t}\right)^{\alpha} = \frac{h_t^{\frac{1+\xi}{\xi}}}{\Gamma_h^{1/\xi}} - \frac{1-\alpha}{\alpha}\frac{\theta}{1+\gamma_0\epsilon} s_t \tau_{t+1},$$

Using this and the discounted pension benefits in equilibrium, we have that

$$\begin{aligned} s_{t,i} =& \frac{1}{1+B_{t+1}} \left[B_{t+1} \frac{\eta_i^{1+\xi}}{X_i^{\xi}} \left(\frac{h_t}{\Gamma_h}\right)^{\frac{1+\xi}{\xi}} \frac{1}{1+\xi} - \frac{1-\alpha}{\alpha}\frac{1-\theta}{1+\gamma_0\epsilon} s_t \tau_{t+1} \right] \\ & - \frac{\eta_i^{1+\xi}}{X_i^{\xi}\Gamma_h} \frac{1-\alpha}{\alpha}\frac{\theta}{1+\gamma_0\epsilon} s_t \tau_{t+1} \end{aligned} \tag{A.4}$$

Using the definition of the aggregate state (1) and rearranging, we then have:

$$s_t = \left(\frac{h_t}{\Gamma_h}\right)^{\frac{1+\xi}{\xi}} \Gamma_{s,t}, \qquad \Gamma_{s,t} \equiv \frac{1}{1+\xi} \frac{\Gamma_h B_{t+1}}{1 + B_{t+1} + \frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}(1+\theta B_{t+1})} \tag{A.5}$$

Furthermore, note that we can combine equations for s_t, h_t to yield closed-form expressions *given* values of B_{t+1}. From the definition of h_t we use that s_t is proportional to $h_t^{(1+\xi)/\xi}$ to derive:

$$h_t = \Theta_{h,t} \left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha\xi}{1+\alpha\xi}}, \qquad \Theta_{h,t} \equiv (\Gamma_h)^{\frac{1+\xi}{1+\alpha\xi}} \left(\frac{(1-\alpha)(1-\tau_t)}{\Gamma_h - \frac{1-\alpha}{\alpha}\frac{\theta\tau_{t+1}}{1+\gamma_0\epsilon}\Gamma_{s,t}}\right)^{\frac{\xi}{1+\alpha\xi}} \tag{A.6}$$

Using this, an alternative expression for s_t is:

$$s_t = \Theta_{s,t} \left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}}, \qquad \Theta_{s,t} \equiv \left(\frac{\Theta_{h,t}}{\Gamma_h}\right)^{\frac{1+\xi}{\xi}} \Gamma_{s,t}. \tag{A.7}$$

Using the relation between s_t and $h_t^{\frac{1+\xi}{\xi}}$ in (A.5), the ratio of individual (A.4) to aggregate

savings satisfies:

$$\frac{s_{t,i}}{s_t} = \frac{\eta_i^{1+\xi}}{X_i^{\xi}} \frac{1}{\Gamma_h} + \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}}{1+\gamma_0\epsilon} \frac{1-\theta}{1+B_{t+1}} \left(\frac{\eta_i^{1+\xi}}{X_i^{\xi}\Gamma_h} - 1 \right). \tag{A.8}$$

For informal households, we assume that informal wages are proportional to formal wages absent taxes and scaled appropriately by informal productivity η_0. Furthermore, we assume that the return to informal savings is proportional to the formal return on savings $R_t^0 = r_0 R_t$.

Specifically, we assume that informal wages are given by:

$$w_t^0 = \left(\frac{1-\alpha}{\Gamma_h^{\alpha}} \right)^{\frac{1}{1+\alpha\xi}} \left(\frac{s_{t-1}}{\nu_t} \right)^{\frac{\alpha}{1+\alpha\xi}}$$

Informal labor supply then follows directly from the derivations above for formal workers; in the case with informal savings, the ratio of informal to aggregate savings is similarly given by

$$\frac{s_{t,0}}{s_t} = \frac{B_{t+1}^0}{1+B_{t+1}^0} \frac{\eta_0^{1+\xi}}{X_0^{\xi}(1+\xi)} \left(\frac{1-\alpha}{\Gamma_h^{\alpha}} \right)^{\frac{1+\xi}{1+\alpha\xi}} \frac{1}{\Theta_{s,t}} - \frac{1}{1+B_{t+1}^0} \frac{1-\alpha}{r_0\alpha} \frac{\epsilon\tau_{t+1}}{1+\gamma_0\epsilon} \tag{A.9}$$

Consumption functions. It is straightforward to derive relevant equilibrium consumption functions $(c_{1,t}^j, \tilde{c}_{1,t}^j, c_{2,t}^j)$ given the results above. Imposing household budget constraints (5), factor prices (3), pension benefits and balanced budget condition (8), and labor and savings decisions from above, we have for formal households i:

$$c_{1,t}^i = \frac{\eta_i^{1+\xi}}{X_i^{\xi}} \frac{s_t}{\Gamma_{s,t}} \left(1 - \frac{B_{t+1}}{(1+B_{t+1})(1+\xi)} \right) + \frac{s_t}{1+B_{t+1}} \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}(1-\theta)}{1+\gamma_0\epsilon} \tag{A.10}$$

$$c_{2,t}^i = \alpha\nu_t h_t^{1-\alpha} \left(\frac{s_{t-1}}{\nu_t} \right)^{\alpha} \left(\frac{s_{t-1,i}}{s_{t-1}} + \frac{1-\alpha}{\alpha} \frac{\tau_t}{1+\gamma_0\epsilon} \left(1 + \theta \left[\frac{\eta_i^{1+\xi}}{X_i^{\xi}} \frac{1}{\Gamma_h} - 1 \right] \right) \right) \tag{A.11}$$

$$\tilde{c}_{1,t}^i = \frac{s_t}{1+B_{t+1}} \left(\frac{\eta_i^{1+\xi}}{X_i^{\xi}} \frac{1}{(1+\xi)\Gamma_{s,t}} + \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}(1-\theta)}{1+\gamma_0\epsilon} \right) \tag{A.12}$$

When informal households save using informal technology, equilibrium consumption functions are given by

$$\tilde{c}_{1,t}^0 = \frac{1}{1+\xi} \frac{\eta_0^{1+\xi}}{X_0^{\xi}} \left(\frac{1-\alpha}{\Gamma_h^{\alpha}} \right)^{\frac{1+\xi}{1+\alpha\xi}} \left(\frac{s_{t-1}}{\nu_t} \right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}} + \frac{1}{1+B_{t+1}^0} \frac{1-\alpha}{r_0\alpha} \frac{\epsilon\tau_{t+1}}{1+\gamma_0\epsilon} s_t \tag{A.13}$$

$$c_{2,t}^0 = r_0\alpha\nu_t \left(\frac{s_{t-1}}{\nu_t} \right)^{\alpha} h_t^{1-\alpha} \left(\frac{s_{t-1,0}}{s_{t-1}} + \frac{1-\alpha}{r_0\alpha} \frac{\epsilon\tau_t}{1+\gamma_0\epsilon} \right). \tag{A.14}$$

When informal households are hand-to-mouth, relevant equilibrium consumption func-

tions are given by[27]

$$\tilde{c}^0_{1,t} = \frac{1}{1+\xi}\frac{\eta_0^{1+\xi}}{X_0^{\xi}}\left(\frac{1-\alpha}{\Gamma_h^{\alpha}}\right)^{\frac{1+\xi}{1+\alpha\xi}}\left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}} \tag{A.15}$$

$$\tilde{c}^0_{2,t} = \frac{1}{1+\xi}\frac{(\chi\eta_0)^{1+\xi}}{X_0^{\xi}}\left(\frac{1-\alpha}{\Gamma_h^{\alpha}}\right)^{\frac{1+\xi}{1+\alpha\xi}}\left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}} + \frac{(1-\alpha)\nu_t\epsilon\tau_t}{1+\gamma_0\epsilon}\left(\frac{s_{t-1}}{\nu_t}\right)^{\alpha} h_t^{1-\alpha}. \tag{A.16}$$

A.2 With log-GHH preferences

With log-preferences, the economic equilibrium is identical to the CRRA equations outlined in appendix A.1, except for the discount function $B_t = B_t^0 = \beta$. All relevant variables $x_t \in \{s_t, h_t, s_{t,i}, h_{t,j}, \tilde{c}^j_{1,t}, c^i_{2,t}, \tilde{c}^0_{2,t}, c^i_{2,t+1}, \tilde{c}^0_{2,t+1}\}$ can be written on the form $x_t = \Theta_{x,t}\left(\frac{s_{t-1}}{\nu_t}\right)^{\sigma_x}$ where σ_x are parameter functions

$$\begin{aligned}
\sigma_h = \sigma_h^j &= \frac{\alpha\xi}{1+\alpha\xi} \\
\sigma^j_{\tilde{c}_1,t} = \sigma^i_{c_2,t} = \sigma^0_{\tilde{c}_2,t} = \sigma_{s,t} = \sigma^i_{s,t} &= \frac{\alpha(1+\xi)}{1+\alpha\xi} \\
\sigma^i_{c_2,t+1} = \sigma^0_{\tilde{c}_2,t+1} &= \left(\frac{\alpha(1+\xi)}{1+\alpha\xi}\right)^2,
\end{aligned}$$

and where the functions $\Theta_{x,t}$ depend only on parameters and policy.

Proof of proposition 1. Part *i.* outlines the marginal effect on s_t, h_t from policy and pension characteristics. Given the solution structure, the effect on s_t, h_t can be found by differentiating equations (A.5) and (A.6) with respect to the relevant variable. For τ_t the effect is straightforward:

$$\begin{aligned}
\frac{\partial \ln(h_t)}{\partial \tau_t} &= -\frac{\xi}{1+\alpha\xi}\frac{1}{1-\tau_t} \leq 0 \\
\frac{\partial \ln(s_t)}{\partial \tau_t} &= -\frac{1+\xi}{1+\alpha\xi}\frac{1}{1-\tau_t} \leq 0.
\end{aligned}$$

We can also show that h_t is unambiguously decreasing in ϵ:

$$\begin{aligned}
\frac{\partial \ln(h_t)}{\partial \epsilon} &= \frac{\xi}{1+\alpha\xi}\frac{\frac{1-\alpha}{\alpha}\frac{\theta\tau_{t+1}}{1+\gamma_0\epsilon}\Gamma_{s,t}}{\Gamma_h - \frac{1-\alpha}{\alpha}\frac{\theta\tau_{t+1}}{1+\gamma_0\epsilon}\Gamma_{s,t}}\frac{1}{1+\gamma_0\epsilon}\left((1+\gamma_0\epsilon)\frac{\partial \ln(\Gamma_{s,t})}{\partial \epsilon} - \gamma_0\right) \leq 0 \\
\frac{\partial \ln(\Gamma_{s,t})}{\partial \epsilon} &= \frac{\gamma_0}{1+\gamma_0\epsilon}\frac{\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}(1+\theta\beta)}{1+\beta+\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}(1+\theta\beta)}.
\end{aligned}$$

[27] Recall informal workers are assumed to receive an endowment proportional to wages of formal workers when they retire. For tractability we define this as if they were making a labor supply decision. The relevant parameter determining the size of the endowment is χ.

However, the effect on s_t is harder to sign.

$$\frac{\partial \ln(s_t)}{\partial \epsilon} = \frac{1+\xi}{\xi}\frac{\partial \ln(h_t)}{\partial \epsilon} + \frac{\partial \ln(\Gamma_{s,t})}{\partial \epsilon}.$$

Replacing from the expressions above, and neglecting common factors, the sign of the derivative is given by the sign of

$$-\frac{1}{1+\alpha\xi}\frac{\theta\beta(1+\beta)}{1+\beta+\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}(1+\theta\beta\frac{\xi}{1+\xi})} + 1 + \theta\beta.$$

This can be seen to be positive for all $\theta \in [0,1]$. Note that the effect from γ_0 is equivalent to that of ϵ. We now consider the effect of future taxes on labor supply and savings.

$$\frac{\partial \ln(h_t)}{\partial \tau_{t+1}} = \frac{\xi}{1+\alpha\xi}\frac{\frac{1-\alpha}{\alpha}\frac{\theta\Gamma_{s,t}}{1+\gamma_0\epsilon}}{\Gamma_h - \frac{1-\alpha}{\alpha}\frac{\theta\tau_{t+1}}{1+\gamma_0\epsilon}\Gamma_{s,t}}\left(1+\tau_{t+1}\frac{\partial \ln(\Gamma_{s,t})}{\partial \tau_{t+1}}\right) \geq 0$$

$$\tau_{t+1}\frac{\partial \ln(\Gamma_{s,t})}{\partial \tau_{t+1}} = -\frac{\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}(1+\theta\beta)}{1+\gamma_0\epsilon}}{1+\beta+\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}(1+\theta\beta)}{1+\gamma_0\epsilon}} \in (-1,0].$$

This shows that labor supply is increasing in τ_{t+1}. For savings the effect is:

$$\frac{\partial \ln(s_t)}{\partial \tau_{t+1}} = \frac{1+\xi}{\xi}\frac{\partial \ln(h_t)}{\partial \tau_{t+1}} + \frac{\partial \ln(\Gamma_{s,t})}{\partial \tau_{t+1}}.$$

As before the sign of this derivative can be shown to be the same as the sign of the following term:

$$\frac{1}{1+\alpha\xi}\frac{\theta\beta(1+\beta)}{1+\beta+\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}(1+\theta\beta\frac{\xi}{1+\xi})} - (1+\theta\beta).$$

This is basically the same term (with different sign) as before. Thus, for all values of θ and τ_{t+1} we have that an increase in τ_{t+1} reduces savings.

Taking the derivative with respect to θ, we see a similar condition as with the effect of τ_{t+1}, namely:

$$\frac{\partial \ln(h_t)}{\partial \theta} = \frac{\xi}{1+\alpha\xi}\frac{\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}\Gamma_{s,t}}{1+\gamma_0\epsilon}}{\Gamma_h - \frac{1-\alpha}{\alpha}\frac{\theta\tau_{t+1}}{1+\gamma_0\epsilon}\Gamma_{s,t}}\left(1+\theta\frac{\partial \ln(\Gamma_{s,t})}{\partial \theta}\right) \geq 0,$$

$$\theta\frac{\partial \ln(\Gamma_{s,t})}{\partial \theta} = -\frac{\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}\theta\beta}{1+\gamma_0\epsilon}}{1+\beta+\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}(1+\theta\beta)}{1+\gamma_0\epsilon}} \in (-1,0].$$

This shows that labor supply is increasing in θ. For the savings rate, we have:

$$\frac{\partial \ln(s_t)}{\partial \theta} = \frac{1+\xi}{\xi}\frac{\partial \ln(h_t)}{\partial \theta} + \frac{\partial \ln(\Gamma_{s,t})}{\partial \theta}.$$

Proceeding as before we find that this is unambiguously negative.[28]

Part *ii.* follows from earlier derivations, specifically equation (A.3). Part *iii.* similarly follows from equation (A.8) with $B_{t+1} = \beta$ in the log case. Finally, part *iv.* follows straightforwardly, comparing (A.8) and (A.3): An average type i household has $\eta_i^{1+\xi}/X_i^{\xi} = \Gamma_h$. Thus, for $\tau_{t+1} = 0$ or $\theta = 1$, we simply have $s_{t,i}/s_t = \eta_i h_{t,i}/h_t$. The spread in savings increases in τ_{t+1} and decreases in θ in the sense that these parameter changes increase $s_{t,i}/s_t$ for $s_{t,i}/s_t > 1$ and decrease for $s_{t,i}/s_t < 1$.

Part *v.* The effects on $s_{t,0}$ and $s_{t,0}/s_t$ from various parameters come from differentiation of (A.9) with $B^0_{t+1} = \beta$. Note that $s_{t,0}$ depends on τ_t only through the effect of taxes on s_t. Thus, it is straightforward to show that both $s_{t,0}$ and $s_{t,0}/s_t$ are increasing in τ_t.

The effects from τ_{t+1} and ϵ are given by:

$$\frac{\partial s_{t,0}}{\partial \tau_{t+1}} = -\frac{1}{1+\beta}\frac{1-\alpha}{r_0\alpha}\frac{s_t\epsilon}{1+\gamma_0\epsilon}\left(1+\tau_{t+1}\frac{\partial \ln(s_t)}{\partial \tau_{t+1}}\right) \leq 0,$$
$$\frac{\partial s_{t,0}}{\partial \epsilon} = -\frac{1}{1+\beta}\frac{1-\alpha}{r_0\alpha}\frac{s_t\tau_{t+1}}{1+\gamma_0\epsilon}\left(\frac{1}{1+\gamma_0\epsilon}+\epsilon\frac{\partial \ln(s_t)}{\partial \epsilon}\right) \leq 0.$$

The first derivative can be signed since $\tau_{t+1}\frac{\partial \ln(s_t)}{\partial \tau_{t+1}} \geq -1$. It is straightforward to show that an increase in θ increases $s_{t,0}$.

To see the effect of τ_{t+1} on the ratio $s_{t,0}/s_t$ we have that:

$$\frac{\partial(s_{t,0}/s_t)}{\partial \tau_{t+1}} = -\frac{\beta}{1+\beta}\frac{\eta_0^{1+\xi}}{X_0^{\xi}(1+\xi)}\left(\frac{1-\alpha}{\Gamma_h^{\alpha}}\right)^{\frac{1+\xi}{1+\alpha\xi}}\frac{1}{\Theta_{s,t}}\frac{\partial \ln(\Theta_{s,t})}{\partial \tau_{t+1}} - \frac{1}{1+\beta}\frac{1-\alpha}{r_0\alpha}\frac{\epsilon}{1+\gamma_0\epsilon},$$

where the derivative $\partial \ln(\Theta_{s,t})/\partial \tau_{t+1} = \partial \ln(s_t)/\partial \tau_{t+1}$ is defined above. We note that this may be negative or positive. For instance, if $\epsilon \approx 0$ an increase in future taxes increases this ratio. Conversely if $r_0 \approx 0$ then we would expect the opposite effect.

B Politico-economic equilibrium

B.1 Proof of proposition 2

As shown before, in equilibrium, all consumption functions are log-separable in the aggregate state s_{t-1}, h_{t-1} (see appendix A.1). In all the Θ-coefficient functions, the only *predetermined* variable that may enter the political problem as an endogenous state is the distribution of savings $s_{t-1,i}/s_{t-1}$. However, the current distribution, $s_{t,i}/s_t$, is

[28] The term whose sign determines this is slightly different

$$\frac{1}{1+\alpha\xi}\frac{1+\beta+\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}}{1+\beta+\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}}{1+\gamma_0\epsilon}(1+\theta\beta\frac{\xi}{1+\xi})} - 1.$$

solely a function of parameters and the *current* policy. Thus, while $\{s_{t-1,i}/s_{t-1}\}_{i>0} \in z_t$, $\{s_{t,i}/s_t\}_{i>0}$ is independent of τ_t implying that

$$\frac{\partial z_{t+1}}{\partial \tau_t} = \frac{\partial \zeta_t}{\partial \tau_t} = 0,$$

where ζ_t is the law of motion for states in the political objective (eq. 9). Thus, we say that this distribution is not an *endogenous* state, as there are no strategic effects from changing τ_t on the future policy τ_{t+1}.

As there are no endogenous states, the politico-economic equilibrium is characterized by the first order condition

$$\omega\left[\gamma_0 \overbrace{\frac{\partial \ln\left(\Theta^0_{\tilde{c}_2,t}\right)}{\partial \tau_t}}^{=\mathcal{E}^0_{2,t}} + \sum_i \gamma_i \overbrace{\frac{\partial \ln\left(\Theta^i_{c_2,t}\right)}{\partial \tau_t}}^{=\mathcal{E}^i_{2,t}}\right]$$
$$+\nu_t\left\{\gamma_0 \underbrace{\left(\frac{\partial \ln\left(\Theta^0_{\tilde{c}_1,t}\right)}{\partial \tau_t} + \beta \frac{\partial \ln\left(\Theta^0_{\tilde{c}_2,t+1}\right)}{\partial \tau_t}\right)}_{=\mathcal{E}^0_{1,t}} + \sum_i \gamma_i \underbrace{\left(\frac{\partial \ln\left(\Theta^i_{\tilde{c}_1,t}\right)}{\partial \tau_t} + \beta \frac{\partial \ln\left(\Theta^i_{c_2,t+1}\right)}{\partial \tau_t}\right)}_{=\mathcal{E}^i_{1,t}}\right\} \leq 0,$$

with equality for an interior solution. We introduce the shorthand notation $\mathcal{E}^j_{x,t}$ to denote the marginal effect of taxes on the indirect utility of generation x of type j. The condition is written for equal propensities to vote, normalized to $\mu_j = 1$. Unequal propensities weight each type's terms by its μ_j. The arguments for ω and ν_t below then go through with the sums so weighted, while those for income inequality and θ, which set the support of high-income retirees against that of low-income ones, need propensities that do not rise too steeply with income: for income inequality the condition is $\mu_H/\mu_L < (c^H_{2,t}/c^L_{2,t})^2$.

Using equilibrium consumption functions from appendix A.1, we have

$$\mathcal{E}^0_{2,t} = \frac{(1-\alpha)\nu_t\epsilon}{1+\gamma_0\epsilon}\Theta^{1-\alpha}_{h,t}\left(1+(1-\alpha)\tau_t\frac{\partial \ln(\Theta_{h,t})}{\partial \tau_t}\right)}{\frac{(\chi\eta_0)^{1+\xi}}{(1+\xi)X_0^\xi}\left(\frac{1-\alpha}{\Gamma_h^\alpha}\right)^{\frac{1+\xi}{1+\alpha\xi}} + \frac{(1-\alpha)\nu_t\epsilon\tau_t}{1+\gamma_0\epsilon}\Theta^{1-\alpha}_{h,t}} \tag{B.1}$$

$$\mathcal{E}^i_{2,t} = (1-\alpha)\frac{\partial \ln(\Theta_{h,t})}{\partial \tau_t} + \frac{\frac{1-\alpha}{\alpha}\frac{1}{1+\gamma_0\epsilon}\left(1+\theta_t\left[\frac{\eta_i^{1+\xi}}{X_i^\xi}\frac{1}{\Gamma_h}-1\right]\right)}{\frac{s_{t-1,i}}{s_{t-1}} + \frac{1-\alpha}{\alpha}\frac{\tau_t}{1+\gamma_0\epsilon}\left(1+\theta_t\left[\frac{\eta_i^{1+\xi}}{X_i^\xi}\frac{1}{\Gamma_h}-1\right]\right)} \tag{B.2}$$

$$\mathcal{E}^0_{1,t} = -\frac{\beta}{1-\tau_t}\frac{\alpha(1+\xi)^2}{(1+\alpha\xi)^2} \tag{B.3}$$

$$\mathcal{E}^i_{1,t} = -\frac{1}{1-\tau_t}\frac{1+\xi}{1+\alpha\xi}(1+\beta), \tag{B.4}$$

It is straightforward to verify that all young households' indirect utility is decreasing

in τ_t. For all retirees, there are opposing effects from adjusting the tax rate: The first term is negative and works through the labor response from current working households ($w_t h_t$ decreases with τ_t) on the pension contributions. The second term is positive and relates to the direct effect of higher pension benefits and the increase in the interest rate.

In an interior equilibrium it must be the case that retirees prefer positive taxes, as the young unambiguously always support zero taxes. Then, increasing ω or decreasing ν_t naturally leads to higher equilibrium tax rates. In the corner with $\tau_t = 0$, there will be no effect from marginal changes in ω, ν_t. Thus $\partial\tau_t/\partial\omega \geq 0$ and $\partial\tau_t/\partial\nu_t \leq 0$. The sign in ν_t needs one more step, since ν_t also scales the universal pension relative to the endowment of informal retirees and so raises their support: with $\sigma_t^0 \equiv b_t^0/\tilde{c}_{2,t}^0$ the pension's share of their consumption, $\partial\mathcal{E}_{2,t}^0/\partial\nu_t = \mathcal{E}_{2,t}^0\,(1-\sigma_t^0)/\nu_t$. Writing $\mathcal{F}_t$ for the left-hand side of the first order condition and using the condition itself to substitute for the young's terms, $\partial\mathcal{F}_t/\partial\nu_t = -\frac{\omega}{\nu_t}\left[\gamma_0\mathcal{E}_{2,t}^0\,\sigma_t^0 + \sum_{i>0}\gamma_i\mathcal{E}_{2,t}^i\right]$, which is negative whenever the aggregate marginal support of formal retirees, $\sum_{i>0}\gamma_i\mathcal{E}_{2,t}^i$, is positive at the optimum, that is, unless the tax is sustained by informal retirees against formal ones.

To analyze the effect of income inequality on equilibrium taxes, without loss of generality we consider two types, H and L, with productivities above and below the average respectively. We raise the productivity of type H and lower that of type L such that

$$\frac{\partial\eta_L}{\partial\eta_H} = -\frac{\gamma_H\left(\frac{\eta_H}{X_H}\right)^{\xi}}{\gamma_L\left(\frac{\eta_L}{X_L}\right)^{\xi}},$$

which keeps the average labor supply and savings (and Γ_h) constant for a given path of policies. For the young generation this does not alter the determinants of taxes, as $\mathcal{E}_{1,t}^i$ is unaffected by it. The effect on retirees $m = L, H$ is given by:

$$\frac{\partial\mathcal{E}_{2,t}^m}{\partial\eta_m} = -(1+\xi)\left(\frac{\eta_m}{X_m}\right)^{\xi}\frac{1}{\Gamma_h}\frac{1-\alpha}{\alpha}\frac{1}{1+\gamma_0\epsilon}\frac{(1-\theta)\left(1+\frac{1-\alpha}{\alpha}\frac{\tau_t}{1+\gamma_0\epsilon}\frac{1}{1+\beta}\right)}{\left(\frac{s_{t-1,m}}{s_{t-1}}+\frac{1-\alpha}{\alpha}\frac{\tau_t}{1+\gamma_0\epsilon}\left(1+\theta_t\left[\frac{\eta_m^{1+\xi}}{X_m^{\xi}}\frac{1}{\Gamma_h}-1\right]\right)\right)^2}$$

This shows that relatively high income households have a weaker preference for taxes. Using this, it is now straightforward to show that $\partial(\gamma_L\mathcal{E}_{2,t}^L + \gamma_H\mathcal{E}_{2,t}^H)/\partial\eta_H > 0$. This is driven by the fact that the marginal utility of consumption is higher for low income households. Thus, higher inequality increases equilibrium taxes.

A similar argument can be applied to see the marginal effects on types H and L of an increase in θ:

$$\frac{\partial\mathcal{E}_{2,t}^m}{\partial\theta} = \frac{\frac{1-\alpha}{\alpha}\frac{\tau_t}{1+\gamma_0\epsilon}\left[\frac{\eta_m^{1+\xi}}{X_m^{\xi}}\frac{1}{\Gamma_h}-1\right]}{\frac{s_{t-1,m}}{s_{t-1}}+\frac{1-\alpha}{\alpha}\frac{\tau_t}{1+\gamma_0\epsilon}\left(1+\theta_t\left[\frac{\eta_m^{1+\xi}}{X_m^{\xi}}\frac{1}{\Gamma_h}-1\right]\right)}\frac{s_{t-1,m}/s_{t-1}}{\frac{s_{t-1,m}}{s_{t-1}}+\frac{1-\alpha}{\alpha}\frac{\tau_t}{1+\gamma_0\epsilon}\left(1+\theta_t\left[\frac{\eta_m^{1+\xi}}{X_m^{\xi}}\frac{1}{\Gamma_h}-1\right]\right)}$$

Thus, an increase in θ increases the support for taxes among the high income households but weakens it among the poor. For the same reasons as above, namely the higher marginal utility of poor households, higher θ then lowers equilibrium tax rates.

The effect of ϵ works through retirees alone, since $\mathcal{E}^0_{1,t}$ and $\mathcal{E}^i_{1,t}$ do not depend on ϵ. Write $\sigma^j_t \equiv b^j_t/c^j_{2,t}$ for the pension's share of the consumption of a retiree of type j, as σ^0_t above, and $\phi_t \equiv 1 + (1-\alpha)\tau_t\,\partial \ln(\Theta_{h,t})/\partial\tau_t$ for the marginal revenue of the tax relative to its base. The expressions above are then $\mathcal{E}^0_{2,t} = \phi_t\sigma^0_t/\tau_t$ and $\mathcal{E}^i_{2,t} = \Big(\phi_t\sigma^i_t - (1-\phi_t)(1-\sigma^i_t)\Big)/\tau_t$: a unit of tax raises every pension by ϕ_t/τ_t in proportion, and with log utility a retiree's support is that proportion times the pension's share of consumption, less, for formal retirees, the fall in the interest rate on their savings. Benefits are shares of revenue $\nu_t w_t h_t \tau_t$, $\epsilon/(1+\gamma_0\epsilon)$ per informal retiree and $(1+\theta_t[y_i-1])/(1+\gamma_0\epsilon)$ per formal retiree of type i with $y_i \equiv \frac{\eta_i^{1+\xi}}{X_i^{\xi}}\frac{1}{\Gamma_h}$, so a higher ϵ raises b^0_t by $1/(\epsilon(1+\gamma_0\epsilon))$ in proportion and lowers every b^i_t by $\gamma_0/(1+\gamma_0\epsilon)$ in proportion, a pure transfer since there are γ_0 informal retirees per formal one. With $s_{t-1,i}/s_{t-1}$ and $\Theta_{h,t}$ held fixed as in the comparative statics above, ϵ enters the first order condition only through the shares, and a share responds to a proportional change in the benefit by $\partial\sigma^j_t/\partial \ln b^j_t = \sigma^j_t(1-\sigma^j_t)$, since the rest of consumption does not move. Hence

$$\frac{\partial \mathcal{F}_t}{\partial \epsilon} = \frac{\omega\,\gamma_0}{\tau_t(1+\gamma_0\epsilon)}\left[\frac{\phi_t}{\epsilon}\,\sigma^0_t\Big(1-\sigma^0_t\Big) - \sum_{i>0}\gamma_i\,\sigma^i_t\Big(1-\sigma^i_t\Big)\right].$$

At an interior equilibrium $\mathcal{F}_t$ is decreasing in τ_t, so taxes rise with ϵ if and only if the bracket is positive,

$$\frac{\phi_t}{\epsilon}\,\sigma^0_t\Big(1-\sigma^0_t\Big) > \sum_{i>0}\gamma_i\,\sigma^i_t\Big(1-\sigma^i_t\Big). \tag{B.5}$$

The share γ_0 multiplies the whole expression, because the loss of formal retirees is the gain of informal retirees times their number, and enters the bracket only through the consumption levels, so it does not determine the sign. The factor $\sigma(1-\sigma)$ is zero for a retiree without a pension and for one who lives on the pension alone, whose consumption and marginal benefit a higher ϵ raises in the same proportion, and is largest at one half. The factor $1/\epsilon$ is the asymmetry of the transfer: a unit of ϵ is a large proportional change of a small universal pension and a small one of the formal pensions. The condition therefore holds when informal retirees are poorer than formal ones and the pension is a comparable share of their consumption, since a comparable share of a smaller consumption is a smaller pension, and fails when they live on the pension alone or are much richer than formal retirees.

B.2 With log-GHH preferences and informal savings

The introduction of $s_{t-1,0}/s_{t-1}$ as an endogenous state variable complicates the model because this state variable not only depends on past choices (through the effect of τ_{t-1} on $\Theta_{s,t-1}$), but is also forward looking ($s_{t-1,0}/s_{t-1}$ depends explicitly on τ_t). To deal with this, let $dx_t/d\tau_t$ denote the effect in politico-economic equilibrium of changing the tax rate τ_t on the variable x_t accounting for the fact that a change in τ_t also changes the state $s_{t,0}/s_t$ which through the continuation policy affects the choice of τ_{t+1} as well. Specifically, for the state variable $s_{t,0}/s_t$ we have:

$$\frac{d(s_{t,0}/s_t)}{d\tau_t} = \frac{\partial(s_{t,0}/s_t)}{\partial\tau_t} + \frac{\partial(s_{t,0}/s_t)}{\partial\tau_{t+1}} \frac{d\tau_t^{+1}}{d(s_{t,0}/s_t)} \frac{d(s_{t,0}/s_t)}{d\tau_t},$$

where the partial derivatives of $s_{t,0}/s_t$ capture how the state reacts to policy in economic equilibrium and $\tau_t^{+1}(s_{t,0}/s_t)$ is the continuation policy that details how policy in $t+1$ adjusts when the endogenous state changes. When considering a marginal change in τ_t, the policymaker takes into account how this propagates via economic equilibrium and continuation policy functions into future periods in order to summarize the total impact on current relevant variables. The technical documentation derives a recursive structure for all the relevant derivatives required to evaluate the marginal effect of τ_t on the indirect utility of households (Berg and Gonzalez-Eiras 2026, section *Informal Savings Model*).

With the endogenous state $s_{t-1,0}/s_{t-1}$, the politico-economic equilibrium is now characterized by the first order condition:

$$\omega\left[\sum_{i\geq 0}\gamma_i \overbrace{\frac{d\ln\left(c_{2,t}^i\right)}{d\tau_t}}^{=\mathcal{E}_{2,t}^i}\right] + \nu_t\left\{\sum_{i\geq 0}\gamma_i \underbrace{\left(\frac{d\ln\left(\tilde{c}_{1,t}^i\right)}{d\tau_t} + \beta\frac{d\ln\left(c_{2,t+1}^i\right)}{d\tau_t}\right)}_{=\mathcal{E}_{1,t}^i}\right\} \leq 0,$$

Using equilibrium consumption functions from appendix A.1, we have the derivatives of indirect utility for retired households:

$$\mathcal{E}_{2,t}^0 = (1-\alpha)\frac{d\ln(h_t)}{d\tau_t} + \frac{\frac{1-\alpha}{r_0\alpha}\frac{\epsilon}{1+\gamma_0\epsilon}}{\frac{s_{t-1,0}}{s_{t-1}} + \frac{1-\alpha}{r_0\alpha}\frac{\epsilon\tau_t}{1+\gamma_0\epsilon}} \tag{B.6}$$

$$\mathcal{E}_{2,t}^i = (1-\alpha)\frac{d\ln(h_t)}{d\tau_t} + \frac{\frac{1-\alpha}{\alpha}\frac{1}{1+\gamma_0\epsilon}\left(1+\theta_t\left[\frac{\eta_i^{1+\xi}}{X_i^\xi}\frac{1}{\Gamma_h}-1\right]\right)}{\frac{s_{t-1,i}}{s_{t-1}} + \frac{1-\alpha}{\alpha}\frac{\tau_t}{1+\gamma_0\epsilon}\left(1+\theta\left[\frac{\eta_i^{1+\xi}}{X_i^\xi}\frac{1}{\Gamma_h}-1\right]\right)}. \tag{B.7}$$

We note that the marginal effect for formal retirees is the same as when continuation policies were independent of state variables. The only difference is that the change in the aggregate labor supply, $\frac{d\ln(h_t)}{d\tau_t}$, now includes strategic effects through future policies.

For the formal, working households, we can use the Euler equation to write the indirect utility as $\ln\left(\tilde{c}^i_{1,t}\right)(1+\beta)+\beta\ln\left(\beta R_{t+1}\right)$. The marginal effect on indirect utility then consists of the two terms:

$$\begin{aligned}
\mathcal{E}^i_{1,t} =&(1+\beta)\frac{d\ln\left(\tilde{c}^i_{1,t}\right)}{d\tau_t}+\beta\frac{d\ln\left(R_{t+1}\right)}{d\tau_t}\\
\frac{d\ln\left(\tilde{c}^i_{1,t}\right)}{d\tau_t} =&\frac{1+\xi}{\xi}\frac{d\ln\left(h_t\right)}{d\tau_t}+\frac{\Gamma_{s,t}\frac{1-\alpha}{\alpha}\frac{1-\theta_{t+1}}{1+\gamma_0\epsilon}\left(\frac{d\tau_{t+1}}{d\tau_t}+\tau_{t+1}\frac{d\ln(\Gamma_{s,t})}{d\tau_t}\right)}{\frac{\eta_i^{1+\xi}}{X_i^{\xi}}\frac{1}{1+\xi}+\Gamma_{s,t}\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}(1-\theta)}{1+\gamma_0\epsilon}}\\
\frac{d\ln\left(R_{t+1}\right)}{d\tau_t} =&(1-\alpha)\left(\frac{d\ln\left(h_{t+1}\right)}{d\ln\left(s_t\right)}-1\right)\frac{d\ln\left(s_t\right)}{d\tau_t}.
\end{aligned}$$

where we use the shorthand $d\tau_{t+1}/d\tau_t = \frac{d\tau_t^{+1}}{d(s_{t,0}/s_t)}\frac{d(s_{t,0}/s_t)}{d\tau_t}$ to define the total effect of a change in the tax rate on future taxes. The technical documentation gives the computation of these derivatives in the numerical application (Berg and Gonzalez-Eiras 2026, section *Informal Savings Model*).

For the informal, working households, a similar Euler condition applies such that we can write indirect utility as $\ln\left(\tilde{c}^0_{1,t}\right)(1+\beta)+\beta\ln\left(\beta R^0_{t+1}\right)$ The marginal effect on indirect utility is then characterized by

$$\begin{aligned}
\mathcal{E}^0_{1,t} =&(1+\beta)\frac{d\ln\left(\tilde{c}^0_{1,t}\right)}{d\tau_t}+\beta\frac{d\ln\left(R^0_{t+1}\right)}{d\tau_t}\\
\frac{d\ln\left(\tilde{c}^0_{1,t}\right)}{d\tau_t} =&\frac{\frac{1-\alpha}{r_0\alpha}\frac{\epsilon}{1+\gamma_0\epsilon}\Theta_{s,t}\left(\frac{d\tau_{t+1}}{d\tau_t}+\tau_{t+1}\frac{d\ln(s_t)}{d\tau_t}\right)}{\frac{\eta_0^{1+\xi}}{X_0^{\xi}}\frac{1}{1+\xi}\left(\frac{1-\alpha}{\Gamma_h^{\alpha}}\right)^{\frac{1+\xi}{1+\alpha\xi}}+\frac{1-\alpha}{r_0\alpha}\frac{\epsilon\tau_{t+1}}{1+\gamma_0\epsilon}\Theta_{s,t}}\\
\frac{d\ln\left(R^0_{t+1}\right)}{d\tau_t} =&(1-\alpha)\left(\frac{d\ln\left(h_{t+1}\right)}{d\ln\left(s_t\right)}-1\right)\frac{d\ln\left(s_t\right)}{d\tau_t}
\end{aligned}$$

B.3 With CRRA-GHH preferences and informal savings

With CRRA preferences it is no longer the case that future policy is independent of the capital stock. But conditional on this state variable, with GHH preferences it is still the case that the distribution of savings for formal workers, $\{s_{t-1,i}/s_{t-1}\}_{i>0}$, remains a function of s_{t-1} and τ_t and is thus not an independent state variable. Thus, the policy function can be written as a function of two endogenous states: s_{t-1} and $s_{t-1,0}/s_{t-1}$.

Compared to the log-GHH case, we adjust the definition of the politico-economic equilibria derivative $dx_t/d\tau_t$ to include the effect through both states such that

$$\frac{dx_t}{d\tau_t}=\frac{\partial x_t}{\partial\tau_t}+\frac{\partial x_t}{\partial\tau_{t+1}}\left(\frac{\partial\tau_t^{+1}}{\partial s_t}\frac{ds_t}{d\tau_t}+\frac{\partial\tau_t^{+1}}{\partial(s_{t,0}/s_t)}\frac{d(s_{t,0}/s_t)}{d\tau_t}\right).$$

The first order condition identifying the politico-economic choice τ_t is now given by

$$\omega \sum_{j\geq 0} \gamma_j \underbrace{\left(c_{2,t}^j\right)^{\frac{\rho-1}{\rho}} \frac{d\ln\left(c_{2,t}^j\right)}{d\tau_t}}_{\equiv \mathcal{E}_{2,t}^j}$$
$$+\nu_t \sum_{j\geq 0} \gamma_j \underbrace{\left(\left(\tilde{c}_{1,t}^j\right)^{\frac{\rho-1}{\rho}} \frac{d\ln\left(\tilde{c}_{1,t}^j\right)}{d\tau_t} + \beta\left(c_{2,t+1}^j\right)^{\frac{\rho-1}{\rho}} \frac{d\ln\left(c_{2,t+1}^j\right)}{d\tau_t}\right)}_{\equiv \mathcal{E}_{1,t}^j} \leq 0,$$

The log-derivatives $d\ln\left(c_{2,t}^j\right)/d\tau_t$ follow from the formulas for the log-GHH case and consumption levels $c_{2,t}^j$ depend on the state s_{t-1} and are given in appendix A.1.

For formal working households, we use the Euler equation to write indirect utility as

$$\begin{aligned}
v_{1,t}^i =& (1+B_{t+1})\frac{\left(\tilde{c}_{1,t}^i\right)^{1-1/\rho}}{1-1/\rho} = \frac{\left(\hat{c}_{1,t}^i\right)^{1-1/\rho}}{1-1/\rho} \\
\hat{c}_{1,t}^i =& \left(\frac{h_t}{\Gamma_h}\right)^{\frac{1+\xi}{\xi}} (1+B_{t+1})^{\frac{1}{\rho-1}} \mathcal{C}_{1,t}^i \\
\mathcal{C}_{1,t}^i =& \frac{\eta_i^{1+\xi}}{X_i^\xi}\frac{1}{1+\xi} + \Gamma_{s,t}\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}(1-\theta)}{1+\gamma_0\epsilon}
\end{aligned}$$

The marginal effect of taxes can then be summed up by

$$\frac{dv_{1,t}^i}{d\tau_t} = \left(\hat{c}_{1,t}^i\right)^{1-1/\rho} \frac{d\ln\left(\hat{c}_{1,t}^i\right)}{d\tau_t} \tag{B.8}$$

$$\begin{aligned}
\frac{d\ln\left(\hat{c}_{1,t}^i\right)}{d\tau_t} =& \frac{1+\xi}{\xi}\frac{d\ln\left(h_t\right)}{d\tau_t} + \frac{B_{t+1}}{1+B_{t+1}}(1-\alpha)\left(\frac{d\ln\left(h_{t+1}\right)}{d\ln\left(s_t\right)} - 1\right)\frac{d\ln\left(s_t\right)}{d\tau_t} \\
&+ \frac{\Gamma_{s,t}\frac{1-\alpha}{\alpha}\frac{1-\theta}{1+\gamma_0\epsilon}\left(\frac{d\tau_{t+1}}{d\tau_t} + \tau_{t+1}\frac{d\ln(\Gamma_{s,t})}{d\tau_t}\right)}{\frac{\eta_i^{1+\xi}}{X_i^\xi}\frac{1}{1+\xi} + \Gamma_{s,t}\frac{1-\alpha}{\alpha}\frac{\tau_{t+1}(1-\theta)}{1+\gamma_0\epsilon}}
\end{aligned} \tag{B.9}$$

For the young informal households, we can employ the Euler equation in a similar way and write:

$$\begin{aligned}
v_{1,t}^0 =& \frac{\left(\tilde{c}_{1,t}^0(1+B_{t+1}^0)^{\frac{\rho}{\rho-1}}\right)^{1-1/\rho}}{1-1/\rho} = \frac{\left(\hat{c}_{1,t}^0\right)^{1-1/\rho}}{1-1/\rho} \\
\hat{c}_{1,t}^0 =& \left(\frac{s_{t-1}}{\nu_t}\right)^{\sigma_s} (1+B_{t+1}^0)^{\frac{1}{\rho-1}} \mathcal{C}_{1,t}^0 \\
\mathcal{C}_{1,t}^0 =& \frac{\eta_0^{1+\xi}}{X_0^\xi}\frac{1}{1+\xi}\left(\frac{1-\alpha}{\Gamma_h^\alpha}\right)^{\frac{1+\xi}{1+\alpha\xi}} + \Theta_{s,t}\frac{1-\alpha}{r_0\alpha}\frac{\tau_{t+1}\epsilon}{1+\gamma_0\epsilon}
\end{aligned}$$

We then have that:

$$\frac{dv^0_{1,t}}{d\tau_t} = \left(\hat{c}^0_{1,t}\right)^{1-1/\rho} \frac{d\ln\left(\hat{c}^0_{1,t}\right)}{d\tau_t} \tag{B.10}$$

$$\begin{aligned}\frac{d\ln\left(\hat{c}^0_{1,t}\right)}{d\tau_t} =& \frac{B^0_{t+1}}{1+B^0_{t+1}}(1-\alpha)\left(\frac{d\ln\left(h_{t+1}\right)}{d\ln\left(s_t\right)}-1\right)\frac{d\ln\left(s_t\right)}{d\tau_t} \\ &+ \frac{\Theta_{s,t}\frac{1-\alpha}{r_0\alpha}\frac{\epsilon}{1+\gamma_0\epsilon}\left(\frac{d\tau_{t+1}}{d\tau_t}+\tau_{t+1}\frac{d\ln(\Theta_{s,t})}{d\tau_t}\right)}{\frac{\eta_0^{1+\xi}}{X_0^{\xi}}\frac{1}{1+\xi}\left(\frac{1-\alpha}{\Gamma_h^{\alpha}}\right)^{\frac{1+\xi}{1+\alpha\xi}}+\Theta_{s,t}\frac{1-\alpha}{r_0\alpha}\frac{\tau_{t+1}\epsilon}{1+\gamma_0\epsilon}}\end{aligned} \tag{B.11}$$

C Endogenous pension design

This appendix collects the derivations behind section 7: the corner that costless redistribution produces, the deadweight cost of redistribution, what the cost changes in the equilibrium of appendices A and B, and the state independence of the design choice under log preferences. Throughout there are no informal households, $\gamma_0 = 0$, so the budget (8) gives $\bar{b}_t = \nu_t w_t h_t \tau_t / h_{t-1}$ and a retiree of type i the benefit $\nu_t w_t h_t \tau_t \left(1 + \theta_t \left[y_i - 1\right]\right)$, where $y_i \equiv \eta_i^{1+\xi}/(X_i^{\xi}\Gamma_h) = \eta_i h_{t-1,i}/h_{t-1}$ is the labor income of type i relative to the mean by (A.3), and $\sum_i \gamma_i y_i = 1$.

C.1 Costless redistribution

We consider three timings of the choice. If θ_t is chosen at t together with τ_t, it re-splits benefits that current retirees have already earned. Current labor supply and savings depend on the design that will apply to the benefits they earn, θ_{t+1}, so the choice moves no economic decision, and its marginal effect on the political objective (14) runs entirely through current retirees,

$$\frac{\partial \mathcal{W}_t}{\partial \theta_t} = \omega \sum_i \gamma_i \mu_i \frac{\frac{1-\alpha}{\alpha}\tau_t\left(y_i - 1\right)}{\frac{s_{t-1,i}}{s_{t-1}} + \frac{1-\alpha}{\alpha}\tau_t\left(1 + \theta_t\left[y_i - 1\right]\right)}. \tag{C.1}$$

The numerator is the change in the benefit of type i and averages to zero across types: a more Bismarckian split moves a given total from retirees below mean income to retirees above it. The denominator is proportional to the consumption of a retiree of type i and rises with income. With equal propensities to vote the sum therefore weights the gains of the rich by a lower marginal utility than the losses of the poor. The following proposition makes the argument exact.

Proposition 3. *Assume log-GHH preferences, no informal households and no deadweight cost of redistribution (section 7.1), and let the design θ_t that applies to the benefits paid at t be chosen at t together with τ_t. Let the propensities to vote be positive and not*

rise with relative income, $\mu_i \leq \mu_j$ whenever $y_i > y_j$, equal propensities included, and let relative incomes not all be equal. Then, for every $\omega > 0$ and $\tau_t > 0$, the derivative (C.1) is negative at every $\theta_t \in [0,1]$ at which every retiree's consumption is positive, and the design chosen together with a positive tax is the Beveridgean corner, $\theta_t = 0$. This holds for all values of α, β, ξ, ν_t, γ_i, η_i and X_i, and whatever tax and design current retirees expected when they saved.

Proof of proposition 3. By (11) the labor supply and the savings of the young at t depend on the design that will apply to their own benefits, θ_{t+1}, and not on θ_t, so θ_t enters no state at $t+1$ and no later decision. It enters $\mathcal{W}_t$ only through the consumption of current retirees (appendix A.1, with $\gamma_0 = 0$),

$$c_{2,t}^i = \alpha \nu_t h_t^{1-\alpha} \left(\frac{s_{t-1}}{\nu_t}\right)^{\alpha} D_i, \qquad D_i \equiv \frac{s_{t-1,i}}{s_{t-1}} + \frac{1-\alpha}{\alpha} \tau_t \left(1 + \theta_t \left[y_i - 1\right]\right), \tag{C.2}$$

where the first factor is common to all types and independent of θ_t. From (14),

$$\frac{\partial \mathcal{W}_t}{\partial \theta_t} = \omega \sum_i \gamma_i \mu_i \frac{\partial \ln c_{2,t}^i}{\partial \theta_t} = \omega \, \frac{1-\alpha}{\alpha} \, \tau_t \sum_i \gamma_i \left(y_i - 1\right) \frac{\mu_i}{D_i},$$

which is (C.1). We sign the sum in three steps.

First, relative incomes average to one. By the definition of Γ_h in (A.3), $\sum_i \gamma_i y_i = 1$, and since $\sum_i \gamma_i = 1$ (section 2.1), $\sum_i \gamma_i \left(y_i - 1\right) = 0$: the design moves a given total between retirees below and above mean income.

Second, D_i rises with y_i. The savings of current retirees were chosen at $t-1$. By part iii of proposition 1, that is (A.8) with $B_{t+1} = \beta$, in which the tax and the design are those that apply to the benefits of the generation that saves,

$$\frac{s_{t-1,i}}{s_{t-1}} = y_i + \frac{1-\alpha}{\alpha} \, \frac{\tau_t^e \left(1 - \theta_t^e\right)}{1+\beta} \left(y_i - 1\right),$$

where τ_t^e and θ_t^e are the tax and the design that generation $t-1$ expected for period t. Whatever these were, the slope in y_i is at least one, since $\tau_t^e \geq 0$ and $\theta_t^e \leq 1$. The second term of D_i has slope $\frac{1-\alpha}{\alpha} \tau_t \theta_t \geq 0$, so D_i rises with y_i with a slope of at least one, and by (C.2) it is positive wherever every retiree's consumption is.

Third, Chebyshev's sum inequality: for weights $\gamma_i \geq 0$ that sum to one,

$$\sum_i \gamma_i a_i b_i - \sum_i \gamma_i a_i \sum_j \gamma_j b_j = \frac{1}{2} \sum_{i,j} \gamma_i \gamma_j \left(a_i - a_j\right)\left(b_i - b_j\right),$$

which is non-positive when a_i and b_i are ordered in opposite directions. Let $a_i = y_i - 1$ and $b_i = \mu_i / D_i$. If $y_i > y_j$, then $\mu_i \leq \mu_j$ and $D_i > D_j > 0$, so $b_i < b_j$, and every term with $y_i \neq y_j$ is negative. Since $\sum_i \gamma_i a_i = 0$ by the first step and relative incomes are not

all equal, $\sum_i \gamma_i (y_i - 1)\, \mu_i / D_i < 0$, and so is the derivative.

Finally, D_i falls with θ_t only for types below mean income, and $D_i \geq 1$ for the others. The designs at which every retiree's consumption is positive therefore form an interval that starts at $\theta_t = 0$ whenever there are any, $\mathcal{W}_t$ is strictly decreasing on it, and the design chosen with any positive tax is $\theta_t = 0$.

The proof uses the propensities to vote only through the ratios μ_i / D_i, which order the types as $\mu_i / c^i_{2,t}$ does. The derivative is therefore negative at every design at which propensities to vote rise with relative income, if at all, by proportionally less than consumption in retirement, $\mu_i / \mu_j < c^i_{2,t} / c^j_{2,t}$ whenever $y_i > y_j$, and a design other than the Beveridgean corner requires propensities to vote that, for some pair of types and at some design, rise at least as fast as consumption. The argument also holds under CRRA-GHH preferences at every intertemporal elasticity: the design still enters only the consumption of current retirees, whose marginal utility is $D_i^{-1/\rho}$ times a factor common to all types, and relative savings keep the form of (A.8) with a discount term B_t common to all types, so $\mu_i D_i^{-1/\rho}$ takes the place of μ_i / D_i.

Propensities to vote that rise with income can overturn the corner, and they do rise in the U.S. calibration of section 6, so there the sign is a quantitative matter: evaluated at the calibrated 2020 equilibrium, (C.1) is negative over the whole interval, at every date of the projection and at $\rho = 0.5$, 1 and 2, and the choice is the corner $\theta_t = 0$.

If instead θ_{t+1} is chosen at t, the design no longer touches the benefits of current retirees but moves current labor supply and savings, directly and through the anticipated response of τ_{t+1}. With the political objective (14) the marginal effect is

$$\frac{\partial \mathcal{W}_t}{\partial \theta_{t+1}} = \sum_i \gamma_i \mu_i \left[\omega\,(1-\alpha) \frac{d \ln \Theta_{h,t}}{d \theta_{t+1}} + \nu_t \left((1+\beta) \frac{d \ln \tilde{c}^i_{1,t}}{d \theta_{t+1}} + \beta \frac{d \ln R_{t+1}}{d \theta_{t+1}} \right) \right], \tag{C.3}$$

where $\Theta_{h,t}$ is the labor supply function (A.6) and each total derivative includes the response of τ_{t+1}, and of θ_{t+2}, to the design chosen. Retirees are unanimous, since their term does not depend on i, and they favour a more earnings-related system: a higher θ_{t+1} raises the return to an hour worked at t and with it the labor supply on which their benefits are levied. The young are divided between this forward-looking stake and the redistribution of their own future benefits, the force of (C.1) one period ahead. The equilibrium is a sequence of policy functions over the inherited design, identified by backward induction as in section 4, and at observed U.S. inequality the redistributive force dominates: the choice is again the corner $\theta = 0$. Section 7 adopts this timing. Of the three, it is the one that keeps the Markov structure and lets the design move labor supply and savings before it binds: a design chosen with the tax moves no decision, and a design chosen once and for all requires a commitment that the Markov equilibria of the paper do not have. It also mirrors how reforms of benefit formulas are legislated, applying to future benefits and leaving benefits in payment untouched.

A design chosen once and for all enters every channel at once. The joint choice of τ_{t_0} and θ is nevertheless one-dimensional, since at any candidate design the first order condition for the tax is the tax condition of appendix B.1 at that design, and the predetermined distribution of savings is held at the value the chosen design implies, which makes the equilibrium a fixed point. The outcome is again a corner: $\theta = 0$ under log preferences, and under CRRA preferences the Bismarckian corner $\theta = 1$ at high intertemporal elasticities, where the forward-looking channels are strong. Near the elasticity at which the corner switches the electorate is close to indifferent between the two, so the switch is not sharply located. The online appendix reports the three timings at each intertemporal elasticity (online appendix); the technical documentation gives the algorithms (Berg and Gonzalez-Eiras 2026).

C.2 The deadweight cost of redistribution

This subsection spells out the argument behind the form of the cost in section 7.1. It motivates the form rather than deriving it from a model of participation and reporting, and it leaves the size of the loss to the calibration.

The implicit tax. A purely earnings-related system that pays out the same revenue returns $\Pi_t \tau_t$ per unit of past earnings to every type, with Π_t the pay-as-you-go return factor of section 7.1. Against that benchmark the flat component levies the implicit tax t_i of (15) on type i: a tax on types above mean income, a subsidy to types below it, and a pure transfer in the aggregate, since $\sum_i \gamma_i y_i t_i = 0$. This is the tax component of the contribution in the sense of Summers (1989), the part of the payroll charge that buys the contributor no benefit, and the only part that distorts like a tax.

Which margin. The wedge (15) is an average implicit tax: it compares what type i receives per unit of its earnings with the benchmark. The marginal wedge is a different object. The flat component depends on aggregate hours and not on own hours, so an extra hour raises the benefit of every type by $\Pi_t \tau_t \theta_t$ per unit of earnings, and the marginal wedge of the flat component, $\Pi_t \tau_t (1 - \theta_t)$, is common to all types. That wedge is already in the model: it is the group $\frac{1-\alpha}{\alpha} \theta \tau_{t+1} \Gamma_{s,t}$ in the labor supply function (A.6), through which hours respond to the design with the Frisch elasticity ξ, and GHH preferences carry no income effect through which the transfers between types could move hours. A cost built on the hours margin would count that response twice and could not depend on the spread of incomes. The cost therefore stands for margins the model does not contain, participation, formality and the reporting of earnings, on which the average wedge is the relevant one: a worker who decides whether to bring earnings into the system compares the earnings at stake with what they buy, so the wedge that matters is the average tax on those earnings (Saez 2002; Kleven and Kreiner 2006). These are the margins on which

the evidence of section 7.1 finds the tax component to matter.

The Harberger loss. A set of proportional wedges t_i on bases e_i, each responding to its net-of-tax rate with a common elasticity ε, costs $\frac{1}{2}\varepsilon \sum_i \gamma_i t_i^2 e_i$ to second order in the wedges (Harberger 1964). Per unit of mean past earnings the bases are $e_i = y_i$, so the loss is $\frac{1}{2}\varepsilon\, \Pi_t^2 \tau_t^2 (1-\theta_t)^2 \tilde{V}$ with

$$\tilde{V} \equiv \sum_i \gamma_i \frac{(y_i - 1)^2}{y_i}, \tag{C.4}$$

while revenue per unit of mean past earnings is $\Pi_t \tau_t$. As a share of revenue the loss is therefore $\frac{1}{2}\varepsilon\, \Pi_t\, \tau_t\, (1-\theta_t)^2\, \tilde{V}$: quadratic in the flat share of the system, linear in its size and proportional to a dispersion of relative incomes that is zero when all incomes are equal. The quadratic counts the subsidy to types below mean income as a loss, as it does the tax on types above it; where participation is already taxed, lowering the participation tax of low-income workers can reduce the total loss (Saez 2002), so the symmetry is a property of the reduced form rather than of the argument.

Specification. The exponential form $f(\theta_t, \tau_t)$ of (16) scales both components of the benefit alike,

$$A(\theta_t, \tau_t) = f(\theta_t, \tau_t)\, \theta_t, \qquad B(\theta_t, \tau_t) = f(\theta_t, \tau_t)\, (1 - \theta_t), \tag{C.5}$$

so that the surviving share of revenue is $A + B = f \in (0, 1]$ and $1 - f$ is, to first order, the Harberger share with λ in the place of $\varepsilon \Pi_t$.[29] The exponential keeps f positive for every λ and every size of system, and nests the model without the cost at $\lambda = 0$. Three properties follow. Nothing is lost when nothing is redistributed, $f = 1$ at $\theta_t = 1$ and for every design when $\tilde{V} = 0$. The loss per unit of redistribution rises with the size of the system, through τ_t. And the marginal cost of the first unit of redistribution is zero, $\partial \ln f / \partial \theta_t = \lambda \tau_t \tilde{V}(1 - \theta_t)$ vanishes at $\theta_t = 1$, so under a purely quadratic loss no electorate chooses the Bismarckian corner, which is why France's design has no own cost (appendix G.1). Scaling both components alike is a choice. A cost on the flat component alone delivers the same endpoints and differs in the interior, but it loses the separate identification of θ and λ (appendix C.3). A cost on the design itself rather than on the transfer it performs, $f(\theta) = \phi + (1 - \phi)\theta^p$, stays in place when a compressed income distribution removes the redistributive stakes; the online appendix reports it.

Magnitude. The shape of f follows from the tax component; its size does not. With Π_t of the order of ν_t, 1.34 in the U.S. calibration for 2020 (table 3), participation and

[29] The return factor multiplies every implicit tax alike and so scales the loss once. It carries ν_t and would make a structural cost parameter drift with demographics; we absorb it into λ, which is calibrated rather than derived.

taxable-income elasticities of 0.1 to 0.4 (Chetty et al. 2011; Saez et al. 2012) correspond to λ between about 0.13 and 0.54. The calibrated values of section 7 are more than an order of magnitude above that range at $\rho \leq 1$, and three to thirteen times above it at $\rho = 2$, so λ also stands in for administrative, evasion and political costs of redistribution that the model does not contain.

C.3 The equilibrium with the cost

With the cost, the benefit formula of (8) becomes

$$b_t^i = \left[A(\theta_t, \tau_t)\, h_{t-1,i}\eta_i + B(\theta_t, \tau_t)\, h_{t-1}\right] \bar{b}_t, \tag{C.6}$$

with $\bar{b}_t = \nu_t w_t h_t \tau_t / h_{t-1}$ the gross revenue per unit of past labor, of which a share $1 - f$ does not reach retirees. The model without the cost is $(A, B) = (\theta_t, 1 - \theta_t)$.

Economic equilibrium. In appendix A the design enters only through the groups $\frac{1-\alpha}{\alpha}\tau_{t+1}\theta_{t+1}$ and $\frac{1-\alpha}{\alpha}\tau_{t+1}(1-\theta_{t+1})$, in $\Gamma_{s,t}$, $\Theta_{h,t}$ and $s_{t,i}/s_t$, and through θ_t and $1 - \theta_t$ in the benefits of current retirees. The cost is therefore exactly the substitution $\theta_{t+1} \mapsto A(\theta_{t+1}, \tau_{t+1})$ and $1 - \theta_{t+1} \mapsto B(\theta_{t+1}, \tau_{t+1})$ in those groups, and $\theta_t \mapsto A(\theta_t, \tau_t)$, $1 - \theta_t \mapsto B(\theta_t, \tau_t)$ in the benefit formula, and it changes nothing else. Under log preferences, (A.5), (A.6) and (A.8) read

$$\Gamma_{s,t} = \frac{1}{1+\xi} \frac{\Gamma_h \beta}{1 + \beta + \frac{1-\alpha}{\alpha}\tau_{t+1}\left[A(\theta_{t+1}, \tau_{t+1})(1+\beta) + B(\theta_{t+1}, \tau_{t+1})\right]}, \tag{C.7a}$$

$$\Theta_{h,t} = (\Gamma_h)^{\frac{1+\xi}{1+\alpha\xi}} \left(\frac{(1-\alpha)(1-\tau_t)}{\Gamma_h - \frac{1-\alpha}{\alpha} A(\theta_{t+1}, \tau_{t+1})\, \tau_{t+1} \Gamma_{s,t}} \right)^{\frac{\xi}{1+\alpha\xi}}, \tag{C.7b}$$

$$\frac{s_{t,i}}{s_t} = y_i + \frac{1-\alpha}{\alpha} \frac{B(\theta_{t+1}, \tau_{t+1})\, \tau_{t+1}}{1+\beta} (y_i - 1), \tag{C.7c}$$

and the consumption of a retiree of type i carries $A(\theta_t, \tau_t)\, y_i + B(\theta_t, \tau_t)$ in place of $1 + \theta_t [y_i - 1]$ in (C.2). The two coincide when $A + B = 1$, so the model without the cost is nested exactly. Here τ_{t+1} is the tax the young anticipate for the period in which their benefits are paid, $\tau_{t+1} = \mathcal{T}_{t+1}(\theta_{t+1})$ with $\mathcal{T}_{t+1}$ the equilibrium tax function below, so the young anticipate both the design and the size of the system that will pay them.

The tax condition. The first order condition for the tax of appendix B.1 picks up one factor. The benefit term of the retirees' marginal effect $\mathcal{E}_{2,t}^i$ is multiplied by τ_t, and f now moves with τ_t, so

$$\mathcal{E}_{2,t}^i = (1-\alpha)\frac{\partial \ln(\Theta_{h,t})}{\partial \tau_t} + \frac{\frac{1-\alpha}{\alpha}\left[A(\theta_t, \tau_t)\, y_i + B(\theta_t, \tau_t)\right]\left(1 + \tau_t \frac{\partial \ln f(\theta_t, \tau_t)}{\partial \tau_t}\right)}{\frac{s_{t-1,i}}{s_{t-1}} + \frac{1-\alpha}{\alpha}\tau_t\left[A(\theta_t, \tau_t)\, y_i + B(\theta_t, \tau_t)\right]}, \tag{C.8}$$

where $\partial \ln f/\partial \tau_t = -\frac{1}{2}\lambda \tilde{V}(1-\theta_t)^2$ and hence $1 + \tau_t\, \partial \ln f/\partial \tau_t = 1 + \ln f(\theta_t, \tau_t)$, since $\ln f$ is linear in τ_t: a larger system is worth less to a retiree at the margin than the revenue it collects, by the share of the marginal contribution that the flat component dissipates. Nothing else in the condition sees f. The term of the young, $\mathcal{E}^i_{1,t}$, contains no policy but τ_t, and $\partial \ln(\Theta_{h,t})/\partial \tau_t = -\frac{\xi}{1+\alpha\xi}\frac{1}{1-\tau_t}$ is free of the groups, so the condition involves θ_{t+1} nowhere and the equilibrium tax at t is a function $\mathcal{T}_t(\theta_t)$ of the inherited design and of parameters and demographics alone, as in proposition 2.

Identification of θ. The replacement rate of type j, its benefit over its past earnings, is $\Pi_t \tau_t \left[A + B/y_j\right]$. The ratio of the replacement rates of two types, the datum that identifies θ in section 6, therefore depends on (A, B) only through $B/A = (1-\theta_t)/\theta_t$, from which f cancels: the design is identified separately from the cost, and $\theta^* = 0.738$ for the U.S. whatever λ. This is a property of the proportional specification. A cost on the flat component alone, $(A, B) = (\theta_t, f(1-\theta_t))$, leaves $B/A = f(1-\theta_t)/\theta_t$, so that the same datum implies a different θ for every value of the cost parameter and the two are identified jointly.

The design condition. With the design chosen one period in advance, θ_{t+1} enters the objective through $A(\theta_{t+1}, \tau_{t+1})$ and $B(\theta_{t+1}, \tau_{t+1})$ in (C.7), with τ_{t+1} the anticipated equilibrium tax. At a given tax, $\partial A/\partial \theta_{t+1} = f + \theta_{t+1}\, \partial f/\partial \theta_{t+1}$ and $\partial B/\partial \theta_{t+1} = (1-\theta_{t+1})\, \partial f/\partial \theta_{t+1} - f$ with $\partial f/\partial \theta_{t+1} = f\lambda\tau_{t+1}\tilde{V}(1-\theta_{t+1})$, so

$$\frac{\partial A}{\partial \theta_{t+1}} + \frac{\partial B}{\partial \theta_{t+1}} = \frac{\partial f}{\partial \theta_{t+1}} = f\,\lambda\,\tau_{t+1}\,\tilde{V}\,(1-\theta_{t+1}) \geq 0, \tag{C.9}$$

zero at $\theta_{t+1} = 1$ and rising with the flat share, the dispersion of incomes and the size of the system the young anticipate. Without the cost a more Bismarckian design shifts next period's benefits one for one from the flat to the earnings-related component, $\partial A/\partial \theta_{t+1} = -\partial B/\partial \theta_{t+1} = 1$, and the redistributive stake of the young dominates as in appendix C.1. With it, a more Bismarckian design also raises the total the young will receive, since it dissipates less, which is what lets the design condition cross zero in the interior.

C.4 The design choice under log preferences

Proposition 4. *Assume log-GHH preferences and no informal households, and let the design be chosen one period in advance under the cost* (C.5)*. Then the political objective at t is additively separable,*

$$\mathcal{W}_t = \mathcal{A}_t\left(\tau_t, \theta_t\right) + \mathcal{B}_t\left(\theta_{t+1}\right) + \mathcal{C}_t\left(s_{t-1}\right), \tag{C.10}$$

where $\mathcal{C}_t$ depends on no policy. The tax chosen at t depends on the inherited design, θ_t, and on neither θ_{t+1} nor s_{t-1}; the design chosen at t maximizes $\mathcal{B}_t$ alone and depends on

neither θ_t *nor* s_{t-1}.

Proof of proposition 4. By (C.7b), $\Theta_{h,t}$ is the product of a factor in τ_t and a factor in $(\tau_{t+1}, \theta_{t+1})$, and $\tau_{t+1} = \mathcal{T}_{t+1}(\theta_{t+1})$ by the tax condition above, so by (A.6) $\ln h_t$ is the sum of a term in τ_t, a term in θ_{t+1} and $\frac{\alpha\xi}{1+\alpha\xi}\ln(s_{t-1}/\nu_t)$. The same holds for $\ln s_t = \frac{1+\xi}{\xi}\ln(h_t/\Gamma_h) + \ln\Gamma_{s,t}$ by (A.5), whose last term is a function of θ_{t+1} alone, and for $\ln\tilde{c}^i_{1,t}$, which by appendix A.1 is $\ln s_t$ plus the logarithm of a bracket in $\Gamma_{s,t}$, θ_{t+1} and τ_{t+1}. The return $\ln R_{t+1}$ is affine in $\ln(s_t/\nu_{t+1})$ and $\ln h_{t+1}$, and $\ln h_{t+1}$ is $\ln\Theta_{h,t+1}$ plus a multiple of $\ln(s_t/\nu_{t+1})$, where $\Theta_{h,t+1}$ depends on τ_{t+1} and on the continuation $(\theta_{t+2}, \tau_{t+2})$; by backward induction from the terminal period, in which no design is chosen, the design chosen at $t+1$ depends on no state, so the continuation is a function of θ_{t+1} alone and every term of the young's utility is a sum of terms in τ_t, in θ_{t+1} and in s_{t-1}. Finally, $\ln c^i_{2,t} = (1-\alpha)\ln h_t + \ln\left[s_{t-1,i}/s_{t-1} + \frac{1-\alpha}{\alpha}\tau_t\left(A(\theta_t,\tau_t)\,y_i + B(\theta_t,\tau_t)\right)\right] + \alpha\ln(s_{t-1}/\nu_t) + \text{const}$, and $s_{t-1,i}/s_{t-1}$ is a function of (τ_t, θ_t) by (C.7c) at $t-1$. Every dependence of (14) on θ_{t+1} therefore arrives additively, apart from the terms in (τ_t, θ_t) and in s_{t-1}, which is (C.10). The first order condition for the tax is $\partial\mathcal{A}_t/\partial\tau_t = 0$, and the design chosen at t is the maximizer of $\mathcal{B}_t$.

The design chosen at t is thus the same whatever system the electorate inherits and whatever the capital stock, which is what section 4 uses: each date's choice is a search over a grid of designs with no state, the tax at each candidate design is found as in proposition 2, and the condition $\theta_{2020} = \theta^*$ that calibrates λ in section 7.2 is a function of λ alone. The property is specific to log preferences. Under CRRA preferences the powers do not separate, both s_{t-1} and θ_t re-enter the design choice, and the equilibrium is a backward recursion over policy functions of the state (s_{t-1}, θ_t) (Berg and Gonzalez-Eiras 2026).

D Terminal states

We consider two different terminal states of the model: A finite horizon version where the economy ends after $t = T$ and a steady state version. The paper reports the finite-horizon solution, in periods of thirty years from 1960 to $T = 2200$ for the rich economies and from 1920 to $T = 2190$ for Argentina, with population growth held at its last projected rate after 2080 (2070 for Argentina).

Finite Horizon. In the finite horizon, consumption functions and indirect utility in economic equilibrium are those of the general case in every period $t < T$. For the young generations at time T, we simply say that

$$u_{T,j} = u_j\left(c^j_{1,T}, h_{T,j}\right).$$

In this case, the economic equilibrium at $t = T$ is the general formulation with $\beta = \tau_{t+1} = s_t = s_{t,j} = 0$. Specifically:

$$h_t = (\Gamma_h)^{\frac{1}{1+\alpha\xi}} \left((1-\alpha)(1-\tau_t)\right)^{\frac{\xi}{1+\alpha\xi}} \left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha\xi}{1+\alpha\xi}} \tag{D.1}$$

$$h_{t,i} = h_t \frac{(\eta_i/X_i)^{\xi}}{\Gamma_h} \tag{D.2}$$

$$\tilde{c}^i_{1,t} = \frac{1}{1+\xi} \frac{\eta_i^{1+\xi}}{X_i^{\xi}} \left(\frac{h_t}{\Gamma_h}\right)^{\frac{1+\xi}{\xi}} \tag{D.3}$$

$$\tilde{c}^0_{1,t} = \frac{1}{1+\xi} \frac{\eta_0^{1+\xi}}{X_0^{\xi}} \left(w_t^0\right)^{1+\xi}. \tag{D.4}$$

For the retired generation, the relevant consumption functions are unchanged.

For the politico-economic equilibrium, the marginal effect of τ_t on retirees is the same as for $t < T$. For working households, the effects are much simpler, namely:

$$\mathcal{E}^i_{1,t} = \left(\tilde{c}^i_{1,t}\right)^{1-1/\rho} \frac{1+\xi}{\xi} \frac{\partial \ln(h_t)}{\partial \tau_t} \tag{D.5}$$

$$\mathcal{E}^0_{1,t} = 0 \tag{D.6}$$

Steady state, infinite horizon. In steady state, we combine equations for aggregate labor supply and savings states to write:

$$s^* = \Gamma_h^{1+\xi} \left[\left(\frac{(1-\alpha)(1-\tau)}{\Gamma_h - \frac{1-\alpha}{\alpha} \frac{\theta\tau}{1+\gamma_0\epsilon} \Gamma_s} \right)^{1+\xi} \frac{\Gamma_s^{1+\alpha\xi}}{\nu^{\alpha(1+\xi)}} \right]^{\frac{1}{1-\alpha}} \tag{D.7}$$

$$\Gamma_s = \frac{1}{1+\xi} \frac{\Gamma_h B}{1 + B + \frac{1-\alpha}{\alpha} \frac{\tau}{1+\gamma_0\epsilon} (1+\theta B)} \tag{D.8}$$

$$B = (\beta)^{\rho} (R^*)^{\rho-1}, \qquad R^* = \alpha \left(\frac{s^*}{\nu h^*}\right)^{\alpha-1}$$

$$\frac{s^*}{\nu h^*} = \frac{(s^*)^{\frac{1}{1+\xi}} \Gamma_s^{\frac{\xi}{1+\xi}}}{\nu \Gamma_h} = \left(\frac{(1-\alpha)(1-\tau)}{\Gamma_h - \frac{1-\alpha}{\alpha} \frac{\theta\tau}{1+\gamma_0\epsilon} \Gamma_s} \frac{\Gamma_s}{\nu} \right)^{\frac{1}{1-\alpha}} \tag{D.9}$$

We can use this to write the discount function as

$$B = (\beta)^{\rho} \left(\alpha \frac{\Gamma_h - \frac{1-\alpha}{\alpha} \frac{\theta\tau}{1+\gamma_0\epsilon} \Gamma_s}{(1-\alpha)(1-\tau)} \frac{\nu}{\Gamma_s} \right)^{\rho-1} \tag{D.10}$$

E Calibration – Argentina

This appendix holds the data, the identifying formulas and the sources behind table 1, which section 5 summarizes.

E.1 The household survey

The Encuesta Permanente de Hogares (EPH) of the Instituto Nacional de Estadística y Censos (INDEC) is Argentina's main household survey of employment, hours and income, and the basis of its official labor market statistics. It is a continuous quarterly survey of the urban agglomerates, 31 metropolitan areas at present, in which households are drawn by a stratified multi-stage design. We use the last two quarters of 2006, a phase of sustained expansion after the recovery from the 2001–2002 crisis, at the start of the expansion of coverage under the amnesties (figure 5.1) and before the Great Recession. The survey asks respondents whether they contribute to social security, which separates formal from informal workers, and we keep those aged 25 to 60.

E.2 The pension system before and after the reform

Before the tax amnesties, retirees over the age of 70 who had made contributions for a minimum of 10 years were eligible for basic pension coverage. This coverage consisted of 70% of the minimum pension and a proportional amount from other contributions.[30] We calculate the pre-reform ϵ as

$$\epsilon = 0.7\left(1-\theta+\theta\frac{\eta_1 h_{t-1,1}}{h_{t-1}}\right)\frac{\beta^{5/30\,\frac{\mathrm{LE}_{\mathrm{men}}-5}{\mathrm{LE}_{\mathrm{men}}}}+\beta^{10/30\,\frac{\mathrm{LE}_{\mathrm{women}}-10}{\mathrm{LE}_{\mathrm{women}}}}}{2},$$

where $\mathrm{LE}_{\mathrm{men}} = 14.45$ and $\mathrm{LE}_{\mathrm{women}} = 22.55$ are the expected life expectancy for men and women, respectively, from their retirement ages (women retire at age 60, men at 65), such that the formula above corrects for the basic pension being received 5 or 10 years after the formal retirement age.[31] We set γ_0 so that informal households are 32% of all households, the share of the population over 65 without a pension before the reform, in 2004 (Rofman and Oliveri 2012); since the formal shares sum to one, $\gamma_0 = 0.32/0.68 = 0.47$. To identify the Bismarckian incentives, we use OECD data on the ratio of replacement rates, RR,

[30] The basic pension was set by Law 24,347 of 1994 at 70% of the first pillar, denoted "prestación básica universal", plus the corresponding fully-funded or pay-as-you-go benefit. For simplicity we assume the latter on average was also 70% of the benefits received by workers that had at least 30 years of contributions. Thus, basic pensions were 70% of the benefits received by the least productive formal workers.

[31] To estimate life expectancy at retirement age for Argentina we conduct a regression analysis using data for men and women's life expectancies at birth and at age 65 for 40 OECD countries for the year 2000. We then use the fitted regression and data from the World Bank on male and female life expectancy at birth in Argentina in 2000.

at half mean and mean wages for males and females.[32] Specifically, we adjust θ to ensure that

$$RR \equiv \frac{\theta + (1-\theta)\frac{h_t}{\eta_3 h_{t,3}}}{\theta + (1-\theta)\frac{h_t}{\eta_1 h_{t,1}}} = \frac{0.678}{0.803},$$

where types 1 and 3, the first and the third quartiles, are those whose mean incomes are closest to half the formal average and to the average, at 0.47 and 1.00 of the average in the survey.

E.3 Technology, demographics, and the aggregate targets

The capital income share is that of the formal sector. In the model, informal households produce outside Y_t and K_t, and national accounts book the output of unincorporated household enterprises, where most informal work sits, as mixed income, which a share computed as one minus the wage bill over GDP counts entirely as capital (Gollin 2002). The economy-wide share for Argentina is 0.43 (Frankema 2010). If a share y of GDP is informal mixed income, the formal-sector share is $(0.43-y)/(1-y)$. Informal earnings of about 5% of GDP, as in the household survey, give 0.40, while shadow-economy estimates of 20–25% of GDP give 0.24 to 0.29. We take $\alpha = 0.35$, which also puts Argentina on the same footing as the Gollin-adjusted shares we use for the rich economies in section 6. We set the Frisch elasticity of labor supply to $\xi = 0.30$. To calibrate β, we target the capital-output ratio. A model period is thirty years and capital depreciates fully between periods, so the capital stock per worker in place at t is the previous generation's savings, $k_t = s_{t-1}/\nu_t$, and with Cobb–Douglas production $K_t/Y_t = \alpha/R_t$ — a stock relative to one period, i.e. thirty years, of output. Thirty times that ratio is the capital-output ratio as the national accounts report it. Targeting it is therefore essentially equivalent to targeting the thirty-year interest rate, which for Argentina is not credibly measured. We target $K/Y = 3.23$, Argentina's ratio in 2010.[33] We calibrate β and ϵ numerically, since β depends on the politico-economic equilibrium path for savings and labor supply and ϵ depends on β.

To construct the time series for ν_t we follow Gonzalez-Eiras and Niepelt (2008) and

[32] In a pure Bismarckian system the replacement rate would be the same at all wage levels, such that $RR = 1$, while in a pure Beveridgean it would double if we halve wages, implying $RR = 0.5$. Note that we combine men and women since we do not have data separately on the increase in coverage or government spending by gender.

[33] Capital stock and GDP at constant national prices (`rnna`, `rgdpna`) from the Penn World Table 11.0 (Feenstra et al. 2015), retrieved through FRED. We read it at the calibration year, where the tax rate, the replacement-rate ratio, the coverage share and the household survey are all measured, rather than averaging. Correcting for the shadow economy would raise the capital-output target, since the Penn World Table stock is economy-wide and the informal sector is labor-intensive. We keep the measured ratio, because no source measures an informal capital share and the stock also includes housing, which cuts the other way.

use 30-year gross population growth rate and projections from census data. Finally, we use social security spending before the 2001 crisis of 7.1% of GDP,[34] to calibrate ω. Since the payroll tax falls on formal labor income, which is a share $1-\alpha$ of formal output, spending of 7.1% of GDP corresponds to an equivalent social security tax rate of $\tau_{2010} = 0.071/(1-\alpha) = 0.109$.

E.4 Productivity and hours

We use η_j and X_j to match the income distribution across formal households and the income and working hours of informal households. Specifically, let z_j^η and z_j^x denote income and hours worked of group j relative to the population-weighted average over the formal households, so that $\sum_{i>0}\gamma_i z_i^\eta = \sum_{i>0}\gamma_i z_i^x = 1$; informal households are measured against the same formal average. With labor supply $h_{t,i}/h_t = (\eta_i/X_i)^\xi / \sum_{j>0}\gamma_j\eta_j^{1+\xi}/X_j^\xi$ for formal households (cf. proposition 1) and $h_{t,0} = (\eta_0 w_t^0/X_0)^\xi$ for informal households, where w_t^0 is the informal wage, we solve the system

$$
\begin{aligned}
z_i^\eta &= \frac{w_t(1-\tau_t)h_{t,i}\eta_i}{w_t(1-\tau_t)h_t} = \frac{\eta_i^{1+\xi}/X_i^\xi}{\sum_{j>0}\gamma_j\eta_j^{1+\xi}/X_j^\xi}, \qquad i>0, \\
z_i^x &= \frac{h_{t,i}}{\sum_{j>0}\gamma_j h_{t,j}} = \frac{(\eta_i/X_i)^\xi}{\sum_{j>0}\gamma_j(\eta_j/X_j)^\xi}, \qquad i>0, \\
z_0^\eta &= \frac{w_t^0 h_{t,0}\eta_0}{w_t(1-\tau_t)h_t}, \qquad z_0^x = \frac{h_{t,0}}{\sum_{j>0}\gamma_j h_{t,j}}.
\end{aligned}
\tag{E.1}
$$

Because only relative income and hours are observed, both the level of efficiency units and the unit in which hours are measured are free. We spend the first on the normalization $\sum_{i>0}\gamma_i\eta_i^{1+\xi}/X_i^\xi = 1$, and take the taste for leisure to be common across the formal groups, $X_i = X$ for $i>0$. The formal income conditions then invert in closed form, $\eta_i = (z_i^\eta)^{1/(1+\xi)}\, X^{\xi/(1+\xi)}$, and they satisfy the normalization at any X, so the level of X is what fixes the unit in which hours are counted. We pin it down by requiring average formal hours in the baseline year, $\sum_{i>0}\gamma_i h_{t,i}$, to equal the average workweek of formal workers in the survey, 42.5 hours expressed as a share of discretionary time. Relative hours across the formal quartiles are then a prediction rather than a target, rising with relative income at elasticity $\xi/(1+\xi)$, and section 5 compares the prediction with the survey. The aggregate h_t weights hours by productivity and is not average formal hours; the two differ by the factor $\sum_{i>0}\gamma_i(\eta_i/X_i)^\xi$, which the informal hours condition in (E.1) carries. That condition and the informal income condition pin down η_0 and X_0 given

[34] Total public pension spending, national and provincial schemes, 7.0% and 7.2% of GDP in 2000 and 2001 (Cetrángolo and Grushka 2020, table 13). We use pre-crisis spending as it was relatively stable during that time, and after 2010 contributive pensions returned to their pre-crisis levels. Following the 2001 crisis, pension spending was reduced, to 5.2% of GDP in 2005, by only allowing minimum pensions to adjust for inflation.

the equilibrium h_t in the baseline year, and we solve for them jointly with β and ω at $X = 1$. The level of X then follows in one step from the solved hours, and η_0 and X_0 are re-derived at it. That last step leaves the equilibrium untouched, since the informal household reaches every aggregate through $\eta_0^{1+\xi}/X_0^{\xi}$, which the unit of hours does not move.

E.5 The vector-X_i calibration

The alternative to a common taste for leisure treats the survey's relative hours as data at every point of the formal distribution: with both z_i^{η} and z_i^{x} imposed, (E.1) identifies one X_i per quartile, and the level of hours is then not identified. Since the productivity distribution reaches every aggregate only through $\sum_{i>0} \gamma_i \eta_i^{1+\xi}/X_i^{\xi}$, which both calibrations normalize to one, β, ω and every equilibrium outcome of section 5, the reform at every ρ and its dependence on ϵ and θ included, are identical under the two; only the productivity and leisure parameters and the reading of relative hours differ. The online appendix reports that calibration against the one used here.

F Calibration – the U.S., France, and the UK

This appendix documents the data and the household heterogeneity behind the calibrations of section 6, whose parameters are in table 3. Each country has three income groups of men aged 25 to 60. The tables report, group by group, the population share γ_i, the taste for leisure X_i and the productivity η_i that the calibration assigns, and the propensity to vote μ_i taken from the data. Under the calibration of section 6 the taste for leisure is common across groups, so the X_i row repeats one number, fixed by the country's average workweek, and the income distribution is carried by η_i alone; the online appendix describes the alternative, with one X_i per group identified from relative hours.

The parameters in the lower panel of table 3 are common to the three countries. The discount factor β matches the U.S. thirty-year interest rate, the gross interest rate net of average multifactor productivity growth estimated by Gonzalez-Eiras and Niepelt (2008), and is imposed on France and the UK. The Frisch elasticity is $\xi = 0.30$, as for Argentina, and the capital share $\alpha = 0.30$.[35] Pension spending is for 2020 (OECD 2021), and ν_t is thirty-year gross population growth from census data and projections, corrected for France as described below; the online appendix reports the calibration at every intertemporal elasticity of the model's grid.

[35] Averaging adjustments 2 and 3 of Gollin (2002) gives capital income shares close to 0.3 for France and the U.S. and slightly lower for the UK.

F.1 The U.S.

For the U.S. we use the CPS March Supplement 2019. The low and medium income groups are cut so that their mean incomes are half the mean and the mean, which is what identifies θ from the replacement rates at those two income levels reported in OECD (2021); the remaining households form the high income group. The propensity to vote is from the U.S. Census, Voting and Registration in the Election of November 2020. To pension spending we add Medicare spending from the 2020 Data Book of the Medicare Payment Advisory Commission, and Medicare enters the calibration of θ as a benefit unrelated to income. France and the UK have no counterpart, since their public health spending is not confined to retirees.[36]

Table 5 shows the two features that set the U.S. apart. Productivity rises steeply across the groups, from 0.82 in the low group to 2.52 in the high one, and the propensity to vote rises with income, from 0.47 to 0.77, so the political objective weighs high-income households well above their population share. The low group holds 58% of households.

Table 5: Household heterogeneity – the U.S.

	Income group			
Parameter	**Low**	**Medium**	**High**	**Target**
γ_i	0.58	0.18	0.24	Income percentiles.
X_i	4.03	4.03	4.03	Average hours worked.
η_i	0.82	1.38	2.52	Income distribution.
μ_i	0.47	0.63	0.77	Voting propensity.

F.2 France

For France we use the 2018 wave of the Luxembourg Income Study. France's system is fully Bismarckian, so $\theta = 1$ whatever the income cuts, and the groups need not be cut at half-mean and mean income. We cut them at the U.S. percentiles instead, which puts the two countries' household types in one-to-one correspondence and makes the counterfactuals of section 6 a type-by-type replacement. The propensity to vote is from the Comparative Study of Electoral Systems.[37] Since the retirement age in France in 2020 was 63.5 years, against 66 in the U.S. and the UK, we take France's population

[36] Bethencourt and Galasso (2008) treat public health care and pensions as political complements, which supports counting Medicare among the transfers to the old that the U.S. electorate votes on.

[37] https://cses.org/.

growth in 2020 to be

$$\frac{\text{Population } 2020 \times (30 - 2.5)/30}{\text{Population } 1990 + \text{Population } 2020 \times 2.5/30},$$

and adjust later years in the same way, with the gap between retirement ages expected to shrink from 2.5 to 2 years.

At the same percentiles France's productivity profile is flatter than the U.S. one (table 6): it runs from 1.12 to 2.56, a ratio of the high to the low group of 2.29 against 3.08 in the U.S. (table 3). The propensity to vote is nearly flat, between 0.81 and 0.86. The common taste for leisure is 7.27 against 4.03 in the U.S., which is how the calibration accounts for France's shorter workweek.

Table 6: Household heterogeneity – France

	Income group			
Parameter	**Low**	**Medium**	**High**	**Target**
γ_i	0.58	0.18	0.24	Income percentiles.
X_i	7.27	7.27	7.27	Average hours worked.
η_i	1.12	1.58	2.56	Income distribution.
μ_i	0.81	0.86	0.85	Voting propensity.

F.3 The UK

For the UK we use the Harmonised British Household Panel Survey, wave 17, which refers to 2006–07 and is distributed by the UK Data Service.[38] Income groups are cut as for the U.S., at half-mean and mean income. The propensity to vote is from the Comparative Study of Electoral Systems.

The UK's cuts put fewer households in the low group than the U.S. cuts do, 29% against 58%, and most of them in the medium group (table 7). The ratio of high to low productivity is close to the U.S. one, 3.06 against 3.08, and the propensity to vote lies between the other two countries, rising from 0.61 to 0.74. Table 10 also reports the UK with its households regrouped at the U.S. income percentiles, by a linear fit of the UK's group means rather than a recount of the survey, which understates the UK's dispersion of relative incomes (appendix G.1); the online appendix documents that grouping.

[38] Later waves were collected at least in part during the Great Recession.

Table 7: Household heterogeneity – UK

	Income group			
Parameter	**Low**	**Medium**	**High**	**Target**
γ_i	0.29	0.58	0.13	Income percentiles.
X_i	8.26	8.26	8.26	Average hours worked.
η_i	0.96	1.63	2.92	Income distribution.
μ_i	0.61	0.70	0.74	Voting propensity.

F.4 France at the UK's income groups

Since France's system is fully Bismarckian, its income groups can also be cut at the UK's own income percentiles without disturbing the identification of θ, which puts France's and the UK's household types in one-to-one correspondence exactly as the U.S. percentiles do above. Table 8 reports France's household heterogeneity on those groups; the online appendix imposes them on the UK.[39] At the UK's cuts France is the more unequal of the two: the productivity of its top group relative to its bottom group is about 3.3 against the UK's 3.06, whereas at the U.S. cuts France is well below the U.S. (table 3). The regrouping moves none of France's aggregates: its political weight of the old and its tax rate are those of table 3.

Table 8: Household heterogeneity – France, UK income groups

	Income group			
Parameter	**Low**	**Medium**	**High**	**Target**
γ_i	0.29	0.58	0.13	Income percentiles.
X_i	7.20	7.20	7.20	Average hours worked.
η_i	0.94	1.51	3.09	Income distribution.
μ_i	0.81	0.83	0.85	Voting propensity.

G Further results on the endogenous design

This appendix collects results on the endogenous design of section 7 that the main text only points to: the calibration of the cost for the UK and the test of the U.S. calibration on the UK and France, and the path of the design along the projected U.S. demography.

[39] Income, hours and group sizes are the same Luxembourg Income Study sample as table 6, regrouped. Voting propensities are only observed at the U.S. cuts; at the UK's cuts each group carries the population-weighted average of the observed values it overlaps with.

G.1 Calibration

Table 9 calibrates the cost on the UK as table 4 does on the U.S.: λ is the value at which the UK electorate re-elects its observed design, $\theta^* = 0.560$, with households grouped at the UK's own income cuts as in section 6. The UK's parameter lies below the U.S. one at low elasticities, by 17% at $\rho = 0.5$ and 16% at $\rho = 1$, and above it at $\rho = 2$, by 61%. The cost weighs less in the UK. At $\rho = 1$ its observed design loses 1.8% of revenue, as the U.S. one does, but a flat benefit of the same size would lose 9.2%, against 23.8% in the U.S., because the UK's system is smaller, with a tax rate of 11.9% against 14.4%, and its dispersion of relative incomes is about half the U.S. one, 0.223 against 0.436. As in the U.S., the cost raises the calibrated political weight of the old, from 1.16 to 1.21 at $\rho = 1$.

Table 9: The calibrated cost of redistribution in the UK

ρ	λ	ω	$1 - f(\theta^*)$	$1 - f(0)$
0.5	15.117	1.76	3.8%	18.1%
1.0	7.264	1.21	1.8%	9.2%
2.0	2.777	1.06	0.7%	3.6%

Note: As table 4, for the UK at its own income groups, where $\theta^* = 0.560$ and $\tilde{V} = 0.223$ at every ρ; λ is the value at which the UK electorate re-elects its observed design. β is the U.S. value of the calibration without the cost, imposed as in section 6, and ω is recalibrated with λ.

Table 10: The UK and France under the U.S. cost of redistribution

Economy	θ^* **observed**	θ **chosen, U.S.** λ	**Own** λ	$\tilde{V}$
UK	0.560	0.625	7.264	0.223
France	1.000	0.749	–	0.218
UK at U.S. income groups	0.543	0.644	6.519	0.065

Note: Each economy is its own calibration at $\rho = 1$ with U.S. $\lambda = 8.643$. "Chosen" is the design in 2020 on the economy's own freely simulated path under the U.S. parameter, to be read against the observed one; "own" is the value at which that path re-elects the observed design. No finite λ makes France re-elect its observed design: it is the corner $\theta = 1$.

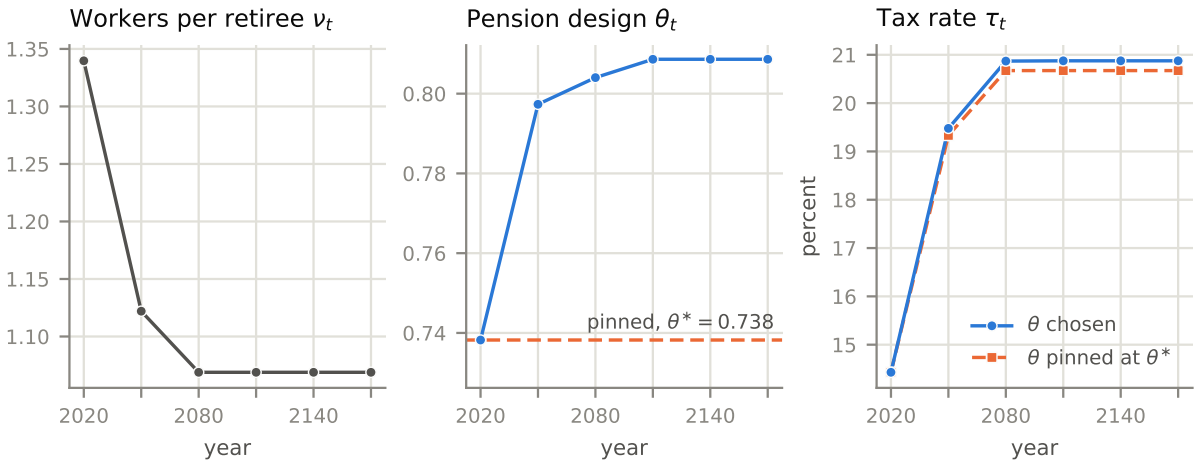
Table 10 asks what the U.S. parameter implies for the other two economies at $\rho = 1$. Each economy keeps its own calibration of section 6 and takes $\lambda = 8.643$ from the U.S. The UK's electorate then chooses 0.625, above its observed 0.560, and France's chooses 0.749, below its observed 1.000, so the three designs are ordered as observed, with the U.S. at 0.738 between them, although France chooses only 0.011 more than the U.S. No finite value of the parameter makes France re-elect its own design, since the cost

does not price the first unit of redistribution (section 7.1). The UK and France have nearly the same dispersion of relative incomes, 0.223 and 0.218, the latter with France grouped at the U.S. income percentiles (about 0.25 at the UK's), while France's tax rate is almost twice the UK's, 21.3% against 11.9%. Under one parameter, each unit of redistribution therefore costs France nearly twice as much as the UK, in line with France's more earnings-related choice. The last row of the table regroups the UK's households at the U.S. income percentiles, by a linear fit of the UK's group means rather than a recount of the survey (appendix F). The regrouping reads the UK's design as 0.543 and its dispersion of relative incomes as 0.065; under the U.S. parameter the regrouped UK chooses 0.644, and its own parameter is 6.519. The linear fit understates this dispersion: the means of the UK's own three groups imply that an exact regrouping at the U.S. percentiles has a dispersion of at least 0.10, so the row is an approximation.

G.2 The design along the baseline path

Figure G.1 follows the U.S. along its projected demography at $\rho = 1$, with the design chosen and with the design pinned at θ^*. Since the calibration targets only the design of 2020, the path from 2020 on is a prediction. Along it the chosen design goes from 0.738 in 2020 to 0.797 in 2050 and stays at 0.809 from 2110 on. It tracks the number of workers per retiree, which falls from 1.34 to 1.07, and settles one period after the demography does. The tax rate climbs from 14.4% to 20.9%, 0.2 p.p. above the path with the design pinned, since a more earnings-related system loses less of its revenue.[40]

Figure G.1: The design and the tax along the baseline path



Note: The paths start in 2020, the calibration year, under log preferences ($\rho = 1$); a design chosen at one date applies from the following one. The online appendix shows the paths from 1990 and for the UK and France, and its page C4 shows them at the other elasticities.

[40]The path barely depends on the Frisch elasticity. Recalibrated at $\xi = 0.2$ and 0.4, λ changes by about one percent and the tax rate in 2110 by about 0.3 p.p., while the design in 2110 and the design under acute ageing are the same to the third decimal (online appendix).